Chapter 8

Valve disease

General principles

Types of valve prosthesis *364*
Homografts and autografts *368*
Choice of valve prosthesis *370*

Aortic valve disease

Pathophysiology of aortic stenosis *372*
Pathophysiology of aortic insufficiency *374*
Timing of aortic valve surgery *376*
Aortic valve replacement: set-up *378*
Valve excision *380*
Aortic valve implantation *382*
Stentless aortic replacement *384*
Homograft aortic replacement *386*
The Ross procedure *388*
Aortic root replacement *390*
Small aortic root *392*
Valve-sparing procedures *394*
Results of aortic valve surgery *396*

Mitral valve disease

Pathophysiology of mitral stenosis *398*
Pathophysiology of mitral regurgitation *400*
Timing of mitral valve surgery *402*
Mitral valve exposure *404*
Mitral valvotomy *406*
Planning mitral valve repair *408*
Principles of mitral valve repair *410*
Mitral valve replacement *412*
Results of mitral surgery *414*

Tricuspid and combined valve surgery

Tricuspid valve surgery *416*
Surgery for infective endocarditis *418*
Combined valvular procedures *420*

Minimally invasive valve surgery is described on 🕮 pp466–467

Types of valve prosthesis

Choice of valve prosthesis is discussed on 📖 pp370–371.

Key facts

- Valve prostheses are either *mechanical or bioprosthetic* ('tissue').
- Tissue valves are mounted on a metal frame (*stented*), or supported by pig aorta and cloth (*stentless*). Stented most commonly used.
- Stented valves are either *porcine* aortic valves or *bovine* pericardium.
- *Homografts* are human cadaveric aortic roots, complete with aortic valves *in situ* (📖 p368).
- A *pulmonary autograft* is the patient's own excised pulmonary valve used in the Ross procedure (📖 p368).

Examples of mechanical valves

Over 50 models of mechanical valves exist, underlining the fact that no one design meets all the criteria of the perfect prosthesis. Examples of three main types are shown in Fig. 8.1:

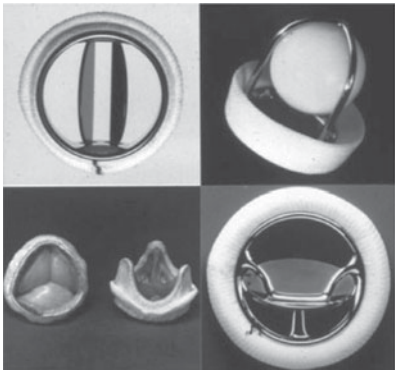
- *Bileaflet valves:* best hemodynamic profile and lowest thromboembolic risk so most widely used today. Commonly inserted models include *CarboMedics* (radio-opaque titanium ring, carbon disks), *St. Jude Medical* (pyrolytic carbon, available with a range of expanded and reduced sewing rings, rotatable and non-rotatable, not radio-opaque).
- *Tilting disk:* largely superseded by bileaflet which have better hemodynamic profile, include *Metronic-Hall A770* carbon coated, disk rotable in sewing ring and *Bjork–Shiley,* carbon coated disk, titanium struts).
- *Ball-in-cage*: First prostheses with long-term durability. Not manufactured now. Higher thromboembolic rate, bulky prostheses. e.g., *Starr–Edwards* best known model.

Examples of tissue valves

There is a similar variety of choices of tissue valves shown in Fig. 8.2. Discovery of glutaraldehyde fixation in the 1960s by Alain Carpentier, followed by antimineralization led to second- and third-generation tissue valves with improved durability. There are two main types:

- *Stented valves:* these are the most commonly implanted, as they are easy to implant being mounted on metal struts with fabric sewing rings
  - *Porcine stented valves:* these are made of pig aortic valves, usually with the porcine non-coronary cusp (which is smaller in the pig) replaced with a coronary cusp to make the valve symmetrical. Examples include *Carpentier–Edwards 2625, Hancock (Medtronic).* The Hancock II has an anticalcification agent added to the fixative. The sewing ring is scalloped to allow supra-annular implantation. *Mosaic (Medtronic):* collagen crimp and flexibility is preserved. So-called because a 'mosaic' of technologies used to increase durability
  - *Bovine stented valves:* these are made of bovine pericardium (equine occasionally used). *Perimount (CE 2700/CE Magna).*

(a)



(b)

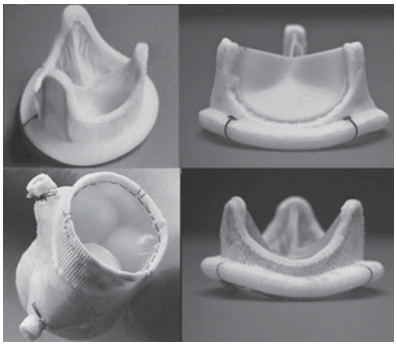


Fig. 8.1 (a) Clockwise from top left: St. Jude bi-leaflet, Starr–Edwards, Bjork–Shiley tilting disk, stented porcine bioprosthesis. Reproduced from Bloomfield P. (2002) Choice of heart valve prosthesis, *Heart*; **87**:583–9 with permission from BMJ Publishing Group Ltd. (b) The four main designs of bioprosthetic valve: porcine intra-annular valve; pericardial intra-annular valve; stentless porcine valve; supra-annular porcine valve. Reproduced from the *ESC Textbook of Cardiovascular Medicine*, with permission from Oxford University Press.

- *Stentless valves:* these are porcine valves supported by porcine aorta. The lack of a stent means they have a better hemodynamic profile, but are technically trickier to implant. The commonest type used is the *Freestyle*, which may be implanted in the subcoronary position as a valve replacement, as an inclusion root cylinder, or as a full aortic root replacement (📖 p390): hence 'freestyle'. *Toronto SPV:* stentless porcine valve. The aortic tissue is removed from all three sinuses and the valve is covered with a thin coat of polyester. It can only be placed in the subcoronary position.

Durability and freedom from reoperation

Figure 8.2 summarizes the main morbidity associated with tissue and mechanical valves, namely rates of thromboembolism and hemorrhage, including stroke, and rates of structural valve degeneration.

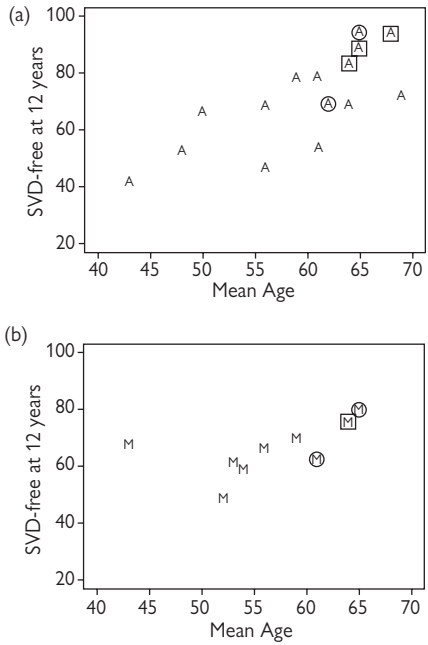


Fig. 8.2 (Continued)

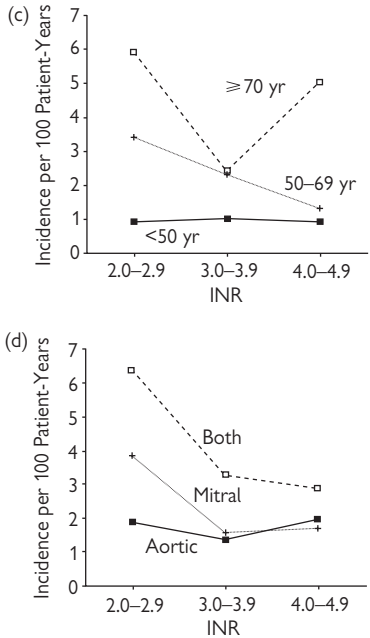


Fig. 8.2 Relationship between freedom from structural valve degeneration at 10 years, patient age, and type of valve implant (a: mitral, b: aortic): squares = pericardial valves, circles = third-generation porcine valves. Note how bioprostheses tend to fail earlier in the mitral position than in the aortic position. This reflects the fact that the mitral prosthesis must remain closed against systolic ventricular pressure, whereas the aortic prosthesis only has to remain closed against aortic diastolic pressure. Valves are thought to fail earlier in younger patients because of their greater hemodynamic demands compared to older patients. (c) INR specific incidence of all adverse events (thromboembolism, major bleeding, stroke) in patients with mechanical valves according to INR and valve position. (d) According to INR range and patient age.

Homografts and autografts

Key facts

- *Homografts (aka allografts)* are aortic roots removed from cadaveric human hearts and implanted as root replacements (📖 p390): the graft does not express MHC antigen as viable endothelial cells disappear within hours of implantation: allograft rejection does not occur.
- *Pulmonary autograft* is the patient's own pulmonary valve, originally implanted as a subcoronary AVR in the Ross procedure but now generally implanted as a root replacement (or very occasionally as a mitral replacement in the Ross II procedure).

Homografts

Homograft procurement

Valves are removed from disease-free cadaveric hearts, unsuitable for transplantation, of donors aged 6 months–55 years, up to 12h after death. The donor heart is procured using aseptic technique, preserved in cold Ringer's lactate solution, and packaged in a sterile plastic bag which is buried in slush.

- Homograft preparation takes place in a clean but not aseptic environment.
- The homograft block consists of all tissue between the base of the LV containing the anterior leaflet of the mitral valve and the aortic annulus, to the innominate artery.
- The coronary ostia are preserved with a small rim of coronary artery.
- The annulus is sized and the length of aorta recorded.

Homograft preservation

There are three main techniques for preservation, which aim to prevent bacterial colonization at the same time as preserving fibroblasts and endothelial cells. Cryopreservation has largely superseded antibiotic preservation. Homovital preservation is a recent development. After harvest the homograft block is stored in a culture medium that contains a low concentration of broad-spectrum antibiotics and stored at 4°C for 24h. The culture medium is changed and the tissue stored for a further 24h at 4°C. It can either be used at this time as a homovital graft or frozen in 10% fetal calf serum and 10% dimethylsulfoxide (DMSO) with controlled rate freezing (fall of 1°C per minute to –40°C) to achieve cryopreservation. This prevents the cellular damage caused by crystallization in conventional freezing processes. Once the homograft reaches –40°C it is transferred to nitrogen storage at approximately –195°C. Bacteriology and serology samples taken prior to storage must be reported clear before the homograft is used. Can be stored for up to 5 years.

Homograft preparation

The homograft is removed from the sterile foil container and placed in RPMI (Roswell Park Memorial Institute solution) with 10% DMSO. By DMSO is removed by serial dilutions and the graft is thawed and rinsed.

Advantages

- Where root replacement is indicated homografts are more pliable than Dacron composite valve replacements, and may be more resistant to

infection (although there is little data to support this) so are favored in surgery for aortic valve endocarditis requiring root replacement where tissue is very friable and reinfection a significant risk.

Disadvantages of homografts

- Homografts are not as durable as mechanical valves, and may not be as durable as porcine or bovine valve replacement.
- Homograft aortic root replacement is a technically demanding operation (📖 p390) and the results are very dependent on the surgeon who requires experience in positioning the homograft in the correct anatomical position, dissecting out the coronary ostia and reimplanting them, and additional suture lines all of which carry an increased risk of postoperative bleeding and prolong the time for which the patient is on bypass, hence national mortality for aortic root replacement is ~7%, compared to ~1–2% for subcoronary AVR.
- Theoretical risk of transmitted infections: TB was transmitted in several cases before preservation and procurement protocols changed.
- Homografts are dependent on donor organ supply, which is falling.
- Homograft tissue is variable in quality, depending on the age of the donor, as well as the skill of the harvester: both the homograft aorta and the valve leaflets may be perforated, or buttonholed, and the tissue can be very friable.

Pulmonary autograft

Using the patient's own pulmonary valve to replace the aortic valve is known as the Ross procedure, after Donald Ross, the British surgeon who pioneered its use. The pulmonary valve is replaced with a pulmonary homograft (Fig. 8.7), which has a life expectancy of 15–20 years. The patient's own pulmonary valve in the aortic position has better life expectancy than bioprostheses, is resistant to infection, does not warrant formal anticoagulation, and most importantly 'grows' with the patient.

Indications for the Ross procedure (controversial)

- Young patients (<21 years).
- Older patients (up to 65 years) in whom an active lifestyle requires good hemodynamic function and contraindicates warfarin, if the surgeon can offer operative mortality <2% (comparable to simple AVR).

Contraindications to the Ross procedure

- Marfan syndrome and other connective tissue disorders.
- Rheumatic valve disease is viewed by some as a contraindication.

Disadvantages of the Ross procedure

The drawbacks are primarily related to the technical difficulty of both procedure and reoperation, and many more surgeons have been discouraged by the steep learning curve and abandoned the operation, than have managed to successfully adopt it as part of their repertoire. Mortality in one meta-analysis ranged from 1 to 7%. Reoperation for autograft dilatation is reduced by modifications where Teflon felt is used to buttress the proximal suture line. This procedure is described on 📖 pp388–389.

Choice of valve prosthesis

Key facts

- There is no strong data showing major long-term survival difference between mechanical vs. tissue valve in either aortic or mitral position.
- In patients >65 years a *bioprosthetic valve* will likely last the patient's lifetime, does not require long-term anticoagulation, and carries a ~1% annual risk of thromboembolic stroke; but valve failure becomes an increasing risk from 10–15 years depending on position and patient age.
- *Mechanical valves* last decades but ~15% will need explanting by 15 years because of endocarditis, pannus, or thrombosis; they require lifelong anticoagulation and have 3–4% *annual* risk of stroke or hemorrhage.

Disadvantages of bioprosthetic valves

Valve failure

Freedom from structural valve deterioration at 12 years in older patients is >90% with third-generation tissue valves such as the Carpentier–Edwards pericardial valves in the aortic position. Tissue valves last longer in the elderly as the hemodynamic demands on the valve are less: the chances of bioprosthetic AVR failure at 12 years is ~40% in patients 0–39 years, ~30% in 40–49 years, and ~10% in >70 years. (Lifetime risk of reoperation similarly highest for younger patients.) Tissue valves last less well in the mitral position compared to the aortic Fig. 8.2, and redo MVR is higher risk than redo AVR: hence the difference in age cut-offs in Table 8.1.

Hemodynamic profile

In smaller sizes (e.g., #19, #21) bioprostheses cause more flow obstruction than same size mechanical valves. *Stented* are more obstructive than *stentless* which are both more obstructive than mechanical.

Table 8.1 Relative indications and contraindications for valve types

	Mechanical	Tissue
Indications	Other mechanical valve (I) >60 years (aortic)* <65 years (mitral)* Other strong indication for anticoagulation	<60 years (aortic)* >65 years (mitral)* Younger patients who wish to avoid (IIa) or have contraindications to anticoagulation (I) Future pregnancy (IIb)
Contra-indications	Inability to comply with anticoagulation (bleeding disorder, social, e.g., IVDU, lack of access) Future pregnancy	Young patient where reoperation would carry high risk of mortality

* In the ACC/AHA guidelines age <65 years is an indication for mechanical AVR: but many feel this should be lower as newer bioprostheses last longer, and reoperation is safer now. I, balance of evidence supports; IIa conflicting evidence balance in favor vs IIb less established.

Calcification

Calcification of bioprostheses occurs in areas of greatest stress: the commissural regions of porcine valves, or the zone of flexion in porcine and pericardial valves, and preludes prosthesis failure. This may happen earlier in patients with renal failure, but the disadvantages of managing anticoagulation in these patients, and their shorter life expectancy, means bioprostheses are a reasonable choice in these patients.

Disadvantages of mechanical valves

Thromboembolic and hemorrhagic events

Without anticoagulation valve life-threatening thrombosis and thromboembolic events are inevitable. Even in an optimally anticoagulated patient (INR 2.5–3) the incidence of major thromboembolic events including stroke and hemorrhage is *still* 2–4% per year (higher in multiple valves and older patients). Bleeding problems (pericardial effusion, late tamponade) are more common postoperatively, and valve thrombosis is a later risk in patients not therapeutically anticoagulated (📖 p267).

Prosthetic valve endocarditis (PVE)

Mechanical valves may be at higher risk of infection than tissue valves during the first 3 postoperative months, but infection rates converge and are similar at 5 years. The cumulative risk of PVE is up to 3.1% at 1 year and 5.7% at 5 years. PVE is a class I indication for urgent valve replacement.

Choice of valve prosthesis in special groups

Younger patients

National registries show a trend towards implanting more tissue valves in younger patients (aged 40–60). This may reflect no long-term survival difference between mechanical and biological valves; lifetime stroke and bleeding rates >60% with mechanical valves; better durability of tissue valves; and fall in redo surgery mortality.

Pregnancy

Pregnant women have increased risk of mechanical valve-related thromboembolism: anticoagulation is vital. Warfarin, a teratogen, is contraindicated in the first trimester, and s/c heparin or i/v heparin should be administered instead, from initial attempts to conceive up until the beginning of the second trimester or delivery. So bioprostheses valves may be implanted in women of childbearing age, who will eventually need redo valve surgery.

Endocarditis

There is no data to suggest choice valve replacement impacts recurrence of endocarditis. Homografts have not been shown to give increased resistance to endocarditis, and they are less durable than porcine valves: their main advantage is being more forgiving to implant in very friable tissue.

Atrial fibrillation

AF is no longer an indication to implant a mechanical valve: concomitant ablation means 60% of patients will be in sinus rhythm at 1 year, obviating need for anticoagulation altogether if a bioprosthesis is implanted.

Small aortic root See 📖 pp392–393.

Pathophysiology of aortic stenosis

Key facts

- Incidence increases with age (rare <40 years of age, 2–3% >80 years).
- Etiology changes with age, although all end up as *calcific* in nature:
  - <70 years: *bicuspid* and *rheumatic* (📖 p78) commonest cause.
  - >70 years: *degenerative* commonest cause.
- Triad of symptoms (syncope, angina, dyspnea) reflect prognosis (Box 8.1).

Morphology

Calcium is deposited in the collagen framework of abnormal leaflets due to the shear stresses occurring with each cardiac contraction. Eventually the valve becomes a lumpy, rigid structure with a tiny orifice, calcification extending down the membranous septum and over the ventricular surface of the anterior mitral valve leaflet. 1–2% of people have a congenitally *bicuspid aortic valve* (right +left cusp fusion in >80%): shear stresses are greater and these valves calcify decades earlier. *Rheumatic disease* causes fibrous leaflet thickening, rolled leaflet edges, and fusion of commissures, and also results in accelerated calcific changes.

Structural and functional changes

The average decrease in the stenotic aortic valve is 0.1cm^2/year. This occurs faster in degenerative and bicuspid stenosis than rheumatic stenosis. Major hemodynamic compromise does not occur until the valve is reduced to less than half the normal aortic valve area of 3–4 cm^2; beyond this LV outlet obstruction rapidly increases. Turbulent flow in the ascending aorta may cause poststenotic dilatation.

- The LV hypertrophies in response to outlet obstruction.
- The number of myocytes is fixed at birth: hypertrophy is an increase in myofibrils, causing existing myocytes to become thicker, not longer.
- There are two types of ventricular hypertrophy: *concentric hypertrophy* which occurs in response to pressure overload, e.g., AS; and *eccentric hypertrophy* which occurs in response to volume overload, e.g., AI. Concentric hypertrophy maintains ventricular volume.
- There is also an increase in interstitial collagen, but not much fibrosis.
- Initially hypertrophy (increase in LV mass from 150 g/m^2 to up to 300g/m^2) allows generation of the high interventricular pressures required to maintain LVEF and CO through the stenosis, despite gradients >100mmHg: this is *compensated aortic stenosis*.
- *Laplace's law* T=Pr/2t: wall stress (T) is proportional to chamber pressure (P) and radius (r), and inversely proportional to wall thickness (t): concentric hypertrophy acts to normalize peak systolic wall stress.
- Eventually hypertrophy cannot compensate for increased wall stress, and the LV thins, dilates and EF falls: *low gradient low EF AS* (📖 p376).

Systolic function

- In compensated AS EF is preserved. Decompensated AS (*low gradient low EF AS*) occurs when systolic function (EF, LV fractional shortening, velocity of shortening) declines as a result of chronic afterload mismatch. *The transvalvular gradient falls* (because cardiac output decreases) *even though the valve stenosis is unchanged or worse* (📖 p376).

Diastolic function

- Diastole has an energy-dependent and a passive component (📖 p60). The passive component depends on normal compliance: this is reduced in AS. The time required for the energy-dependent phase is greater because of impaired myocardial relaxation, and decreased chamber compliance. LVEDP is increased, reducing diastolic coronary flow (📖 p62). Adequate LV filling is very dependent on '*atrial kick*' and controlled rate (📖 p60). Patients may: (1) exhibit diastolic failure with PND, orthopnea and pulmonary edema even with normal EF, (2) decompensate if they go into AF, and (3) often take months to notice a symptomatic benefit after surgery

Coronary blood flow

In AS the myocardial oxygen supply–demand mismatch causes angina even in patients with normal coronary arteries, because of:

- Increased myocardial oxygen demand (increased muscle mass).
- Which outpaces compensatory increase in blood flow resulting in lower coronary blood flow per 100g of myocardium.
- ↑↑LVEDP reduces coronary perfusion during diastole.
- Increased systolic time reduces diastolic filling time.
- Decreased coronary vasodilatory reserve results in decreased coronary flow in exercise.
- Peripheral exercise induced vasodilatation reduces coronary perfusion.

Symptoms

Patients may be asymptomatic. Most present with any of the triad of angina, exertional syncope, and dyspnea. There is a 5% incidence of sudden death. Symptom onset is closely related to survival.

Box 8.1 Prognosis is predicted by symptoms in AS

- *Angina:* 50% of patients dead within 5 years without AVR, earliest symptom, due to O_2 supply–demand mismatch.
- *Exertional syncope:* 50% of patients dead within 3 years without AVR—due to combination of a relatively fixed CO which cannot respond to the drop in systemic perfusion pressures caused by exercise-induced peripheral vasodilatation.
- *Dyspnea:* 50% of patients dead within 2 years without AVR, late sign due to diastolic heart failure and/or congestive heart failure.
- *Sudden death* (5%) may be due to the increase in ectopy caused by impaired oxygen supply and changes at the cellular level, and complete heart block caused by calcification of conducting pathways.

Signs

- Slow rising, low amplitude pulse, occasionally with thrill at carotids.
- Sustained apex beat.
- Soft first heart sound, sometimes preceded by fourth heart sound, single heart sound if valve is calcified, ejection systolic murmur radiating from aortic area to right carotid.
- Soft early diastolic murmur (AI as leaflets become immobile).

Pathophysiology of aortic insufficiency

Key facts

- Commonest cause of isolated severe AI in West is aortic root/annular dilatation, followed by bicuspid valve, infective endocarditis, and rheumatic disease (📖 pp78–79).
- Acute severe AI is a surgical emergency.
- Chronic AI may be asymptomatic for years: symptoms include angina, exertional dyspnoea, and eventually cardiac failure.

Morphology

Pathology of any of the components of the aortic root can prevent leaflet coaptation, leading to AI:

- Annular dilatation (commonest cause in West), e.g., annuloaortic ectasia caused by cystic medial dilatation associated with connective tissue disorders such as Marfan syndrome, type A aortic dissection.
- Leaflet thickening/shortening due to rheumatic disease.
- Leaflet perforation (endocarditis).
- Calcification causing AS may also cause AI due to decreased leaflet mobility.

Structural and functional changes

Acute aortic insufficiency

Most commonly caused by endocarditis, aortic dissection, and trauma, this is a surgical emergency. There is no time for any of the adaptive changes that characterize chronic AI to develop and the sudden volume overload results in pulmonary edema as the filling pressure is increased in a non-compliant ventricle. Reduced coronary blood flow, low cardiac output, and hypoxia result in end-organ ischemia.

Chronic aortic insufficiency

Chronic AI is better tolerated than acute partly because of the following adaptive and maladaptive changes:

- AI leads to both volume overload, because of the regurgitant fraction, and pressure overload, because of the increased volume of blood ejected.
- Volume overload causes eccentric hypertrophy and LV dilatation.
- Pressure overload results in concentric hypertrophy: the same increase in myofibrils and interstitial collagen described in AS occurs in AI.
- Ventricular enlargement produces a larger total stroke volume (up to 20L/min), all of which enters the aorta during systole: this is compensated AI and the resultant increase in pulse pressure causes systemic hypertension and a large increase in LV afterload.
- LV compliance decreases and diastolic function is compromised as hypertrophy progresses.
- Eventually LV end-diastolic pressures and LAPs begin to increase: pulmonary hypertension and congestive cardiac failure is a feature of decompensated AI as well as acute AI.
- Finally as myocytes are stretched beyond their limit of effective contraction LV systolic function decreases, and the rate of increase LV end-systolic diameter accelerates to about 7mm per year.
- Coronary perfusion may be reduced for the same reasons as in AS.

Symptoms

Acute AI is characterized by cardiovascular collapse and the symptoms of pulmonary edema. Chronic AI is usually asymptomatic until LV function is significantly impaired. Symptoms eventually include angina (because of myocardial O_2 supply–demand mismatch caused by combination of increased O_2 consumption due to ventricular hypertrophy and contractility, and decreased coronary reserve caused by the fall in aortic–LV diastolic pressure gradient), effort dyspnea, and the symptoms of congestive cardiac failure. Fatigue may predominate.

Signs

- Large amplitude collapsing or 'water-hammer' pulse with a rapid upstroke.
- Quinke's sign: capillary pulsation visible in the nail bed.
- Corrigan's sign: visible pulsation in the neck.
- De Musset's sign: head nodding as a result of arterial pulsation.
- Duroziez's sign: 'pistol-shot' femorals.
- Sustained, laterally displaced apex beat.
- Single second heart sound, with third heart sound.
- Early diastolic murmur maximal at lower left sternal edge radiating to the apex and axilla.
- Ejection systolic murmur may be present.
- Late diastolic murmur (Austin Flint) from fluttering of the anterior mitral leaflet as it is hit by the regurgitant jet.

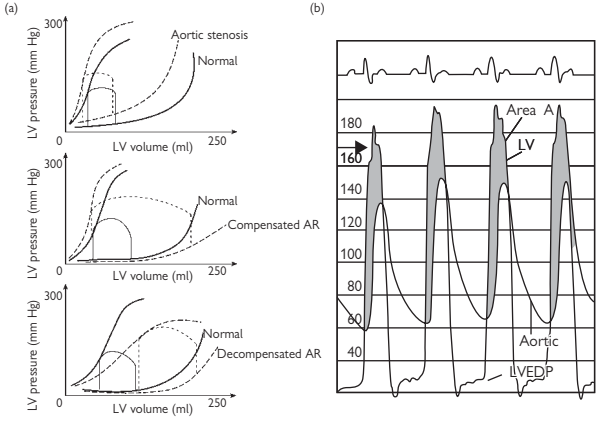


Fig. 8.3 (a) Pressure–volume loops in compensated AS and AI, and decompensated disease. (b) Cardiac catheterization tracing in aortic stenosis: note the difference in LV and aortic pressure which corresponds to the peak transvalvular gradient.

Timing of aortic valve surgery

Key facts

- Severe AS is defined as an aortic valve area (AVA) <1cm^2.
- If mean gradient >40mmHg then AS is severe, but *severe AS may be present at lower gradients* if the patient has poor LV function.
- The presence of symptoms or LV systolic dysfunction are both class I indications for AVR in both AS and AI.

Symptomatic aortic stenosis

The onset of symptoms in patients with AS is a poor prognostic sign (see Box 8.2). There is a 5% incidence of sudden death per year. Symptoms are most likely due to AS if AVA <1cm^2 or the transvalvular mean gradient >40mHg, and symptom onset in these patients mandates prompt AVR in all except very high-risk cases.

Low gradient, low EF (low flow) AS

There is a subgroup of symptomatic patients with transvalvular gradients of <40mmHg which fall into one of three main groups:

- Low gradient, low EF AS: severe AS results in ⇊myocardial contractility with low gradient due to low cardiac output. Dobutamine infusion during echo or cardiac cath can help identify high risk candidates:
  - Positive *dobutamine stress test*: if cardiac output and transvalvular gradient increase in response to dobutamine, it suggests there is

Box 8.2 Indications for aortic valve surgery

Class I indications for AVR in AS

- Symptomatic severe AS (AVA <1.0cm^2 or mean gradient >50mmHg).
- Patients with moderate AS undergoing other cardiac surgery.
- Asymptomatic patients with severe AS and LV systolic dysfunction.

Class IIb indications for AVR in AS

- Asymptomatic AS and abnormal response to exercise (e.g., ↓BP, symptoms), or AVA <0.6cm^2/VT, LV hypertrophy>/=15mm, very high gradient or calcification/CAD implying rapid progression risk.

Class I indications for AVR in AI

- Acute severe AI, symptomatic severe AI.
- Severe AI undergoing CABG or other valve surgery.
- Severe asymptomatic AI and LVEF <50%.

Class IIa indications for AVR in AI

- Severe AI and LVESD >55mm or LVEDD >75mm.

Class IIb indications for AVR in AI

- Severe AI with normal EF and LVESD >50mm or LVEDD >70mm *and* progressive dilatation or declining exercise tolerance.

Indications for AVR in endocarditis (see 🕮 p418 for details)

- Endocarditis and AI/AS resulting in heart failure, ↑LVEDP, LAP, or PAP (I), heart block, annular abscess, destructive penetrating lesion (I), recurrent emboli and vegetation despite appropriate antibiotics, fungal or other virulent organism (IIa), vegetation >10mm (IIb).

a true flow-limiting lesion in the setting of reversible ventricular dysfunction: surgery beneficial although higher risk than normal LV.
  - In the irreversibly damaged LV there is no increase in transvalvular flow in response to inotropes, and (lack of *contractile reserve*) these patients present with high post-op mortality and worse survival.
- Pseudo AS: severe cardiomyopathy due to non-valvular cause, with only moderate or mild AS. The AVA increases during dobutamine stress testing. No benefit from surgery. CT or MRI can accurately quantify AVA and define abnormal myocardium.

Asymptomatic aortic stenosis

The majority of asymptomatic patients with severe AS have near normal life expectancy without AVR. Their 1–2% incidence of sudden death is less than the mortality of AVR, making it crucial to select only those patients at high risk of sudden death without surgery. Two techniques are employed: echocardiography and exercise testing. Asymptomatic patients with a transvalvular gradient >64mmHg have a 80% chance of becoming symptomatic within 2 years (hence an ACC/AHA IIb indication if surgery low risk). Exercise testing safe in asymptomatic patients, and can identify the presence of exercise-induced hemodynamic compromise which is an ACC/AHA IIb indication for AVR. The natural history of AS in asymptomatic patients with a positive exercise test is probably very similar to that of patients with exertional symptoms.

Aortic insufficiency

Acute severe AI is a surgical emergency. In infective endocarditis with lesser degrees of AI, the indications for early operation are annular abscess formation, new heart block, or evidence of heart failure (📖 p418).

Chronic aortic insufficiency

Even in severe AI it may be 10 years before symptoms become severe. The prognosis during this period is good: in severe chronic AI with preserved ejection fraction has an 80% 5-year survival. Surgery should take place before permanent LV dysfunction ensues: so surgery is often indicated in asymptomatic patients. Progressive LV enlargement or ↓LVEF heralds onset of symptoms and decompensation. AVR should therefore be carried out before LVEF <50%, LVESD >55mm, or LVEDD >70mm.

Elderly patients

The normal life expectancy for octogenarians is listed in Table 8.2. AVR in selected octogenarians has <5% operative mortality, and results in improved quality and length of life in patients with symptomatic AS.

Table 8.2 Normal life expectancy in octogenarians

	Life expectancy	
Age (years)	**Males (years)**	**(Females (years)**
70–75	12.2	15.3
75–80	9.5	11.9
80–85	7.1	8.9

Aortic valve replacement: set-up

Before you scrub

Note whether the coronary circulation is left or right dominant, and whether the left main stem is short. Is there AI (see below)? What is the size of the aortic root—small (📖 p392), aneurysmal (📖 p444).

Set-up

The patient is positioned and draped in the usual manner. Usually CO_2 insufflation on field, myocardial temperature probe, insulating pad.

Bypass

Standard cardiopulmonary bypass (CPB) at 32°C, two-stage venous cannula and standard aortic cannula (as distal as possible to provide adequate space for cross-clamp, aortic root vent, aortotomy, and proximal coronary anastomoses if planned). 4/0 Prolene purse-string for RSPV vent. Separate the PA from the aorta. Some surgeons tape the aorta.

Cardioplegia

Give anterograde cardioplegia to arrest the heart via a Medicut cannula: once the aortic root is opened further doses of cardioplegia are given with a handheld cannula (size 6 or 7 for LM ostia, size 5 or 5 balloon tip for RCA ostia) directly into each coronary sinus, and/or retrogradely.

- If there is AI the options are: (a) attempt to arrest the heart as described—can often obtain a good root pressure by lifting up on clamp and giving plegia rapidly despite AI—just need to ensure LV well vented so it does not distend; (b) fibrillate the heart on bypass, perform the aortotomy and give plegia directly into the coronary ostia; or (c) site a retrograde cardoplegia cannula and arrest heart with retrograde (takes a little longer). Retrograde plegia is useful routinely for AVR because:
  - Coronary ostia occasionally too difficult to cannulate directly.
  - Can continue to operate while giving retrograde cardioplegia.
- If the LMS is short be very careful not to push the handheld cardioplegia too far into the coronary ostia, as there is a risk of only protecting the LAD or the circumflex territory and not both.
- If the coronary circulation is right dominant give a larger amount of plegia into the right ostia, otherwise ratio of left:right plegia is ~2:1.

Venting

Any of the following routes can be used to vent the heart:

- Pump sucker placed retrogradely into the LV via the aortotomy (easy to place once aortotomy made but gets in the way a bit).
- LV apical vent (easy to place, but less easy to guarantee hemostasis).
- RSPV vent (sometimes less easy to place into LV, but out of the way and easy to secure hemostasis). Venting the PA also works.
- For de-airing it is safest to use aortic root vent until TEE free of air, rather than de-airing through the last suture of aortotomy closure.
- Prosthetic valves need the heart to eject to open and close: if you remove cross clamp before the heart has a rhythm the LV will distend:
  - Give hot-shot for 5–10min before you are ready to remove clamp.

- *Ensure* your LV vent is in place before starting aortomy closure, especially if patient cooled as may take long time to resume rhythm.
- Place an epicardial pacing wire before you remove the clamp.

Aortotomy

The goal of the aortotomy is to get close enough to the aortic valve to ensure good access, without getting so close that the RCA ostia and closure of the aortotomy are compromised. The choice of incision is dictated by any planned procedure on ascending aorta, the choice of prosthesis, the need for aortic root procedures such as enlargement, and concomitant CABG.

- Retract the fad pad overlying the aortic root.
- Separate the aorta from the PA, and carefully free the fat off the aorta proximally, taking care not to damage the RCA which arises close by.
- The standard aortotomy is made starting about 1cm from the sinotubular junction (~2 cm cephalad to the origin of the right coronary artery) starting at the uppermost point of the ascending aorta and extending one end of it proximally down towards the middle of the non-coronary cusp at the sinotubular junction, and the other end transversely (Fig. 8.4).
  - This incision permits aortic root enlargement and affords maximal exposure of the aortic valve (📖 pp392–393).
  - If in doubt make your aortotomy more distal rather than too proximal to avoid damaging the RCA ostia.
  - If you take the aortotomy too far into the NCC, it is more challenging to close it around the strut of a stented prosthesis, whereas if you do not take it far enough annular suture placement is a bit more difficult as you are effectively operating down a tunnel.
- A transverse aortotomy centred 15mm above the sinotubular junction extending round 2/3 of the aortic circumference gives adequate access to the aortic root without distortion: it is not easily adapted to aortic root enlargement.
- The aorta is transected about 15mm from the sinotubular junction (Fig. 8.4) which provides good exposure of the aortic root, but bleeding from the posterior aspect of the aortotomy suture line may be problematic: this incision is normally reserved for root replacement (📖 p390).

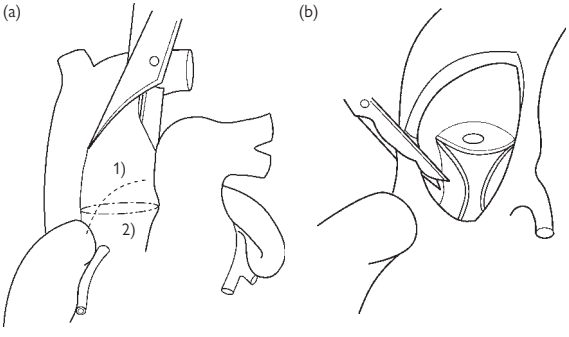


Fig. 8.4 (a) Oblique aortotomy and transverse aortotomy. (b) Extending the aortotomy.

Valve excision

Valve excision

The aortotomy is kept open using a Ross retractor or stay sutures. The first assistant's main role is to prevent stroke and perioperative myocardial infarction by ensuring all debris is removed using rough sucker with the guard removed. LV vent suction keeps the operative field blood-free.

- Use the McIndoe scissors to excise the valve leaflets by cutting along the edge of attachment: easiest to start at the non-coronary (NCC) and right coronary cusp (RCC).
- Scissors are used because the blunt tips will not perforate the aorta.
- Once the valve leaflets are excised use a rongeur, or a bone nibbler to remove calcium extending onto the aortic wall, the ventricular septum, and anterior mitral leaflet.
- Grasp each piece of calcium and gently remove with a twisting motion, using an 11 blade or McIndoe scissors to divide fibrous tissue.
- Leaving calcium predisposes to paravalvular leaks.
- Damage to the interventricular septum results in complete heart block.
- The area between the left and the NCC requires particular care, as it easy to perforate the aorta or the junction between the aorta and the ventricle here: if this happens the defect should be repaired before the valve is inserted.
- After the valve is excised run a finger round the inside of the annulus to detect residual calcium.
- Turn off vents and irrigate with cold saline which washes out any debris and affords topical cooling.

Sizing and preparing the valve

- Use the same make of sizer as the type of valve to be inserted: sizes vary between makes. The correct sizer is the largest one which passes with a bit of resistance into the LV. Always compare this with the sizes above and below. See 📖 p392 for patient–prosthesis mismatch (PPM).
- Although optimal aortic valve area depends on body surface area (📖 p392), a good rule of thumb is that valve sizes of 23mm and above offer good hemodynamic performance: valves of 19mm should only be inserted in small, sedentary adults: otherwise the intrinsic gradient is high.
- If the aortic root is small, forcing an oversized prosthesis into position will result in damage to the aorta and erosion of the annulus. There are several options (📖 p392). A larger prosthesis can be placed in the supra-annular position than in the annular position (Fig. 8.5), alternatively the valve can be inserted at a slight angle, by suturing it in the annular position along the right and left coronary cusps, and suprannularly along the non-coronary (Fig. 8.5b).
- Bioprostheses need 3 × 2min washes in saline to remove glutaraldehyde solution: this is done while placing annular sutures.

Valve implantation

- Valve implantation is described on 📖 pp382–383. Fig. 8.5b shows the relative positions of subcoronary annular and suprannular implantation, together with schema of cylinder and root replacement. Retrograde cardioplegia can be given while placing the sutures in the valve ring.

Closing the aortotomy

The commonest site of bleeding is at either end of the aortotomy, caused by either unrecognized tears in the aorta or incomplete closure. Avoid this by starting with a pledgetted mattress suture 2–3mm beyond the aortotomy at either end using 4/0 Prolene, and closing the aorta in one or two layers, using a simple running stitch meeting in the middle or horizontal mattress sutures which may be pledgetted (if fragile or very thin). Take care not to catch the back wall of the aorta (easy to do if the cross-clamp is close to the aortotomy) or the valve mechanism.

- Switch off the vents midway through the aortotomy closure otherwise the heart will be full of air by the time you finish.
- In small aortas the valve struts may prevent primary closure: use a diamond-shaped patch of pericardium and a 4/0 Prolene suture to close the defect a *patch aortoplasty* with several interrupted horizontal mattress sutures in the corners, and then running suture.

De-airing and evaluating valve

- Place the patient in Trendelenburg. de-airing is described on 📖 p114.
- Assess TEE for residual AI, gradient, prosthesis function, other valves:
  - AI may be due to 'wash-jets' usually two parallel small jets normally seen with bileaflet valves; of no physiological consequence.
  - Paravalvular leaks due to suture holes may disappear after protamine, larger ones may need revision (📖 p266).
  - Moderate gradient across small prosthesis common (📖 p392).
  - New MR from aortomitral curtain if overly aggressive decalcification.

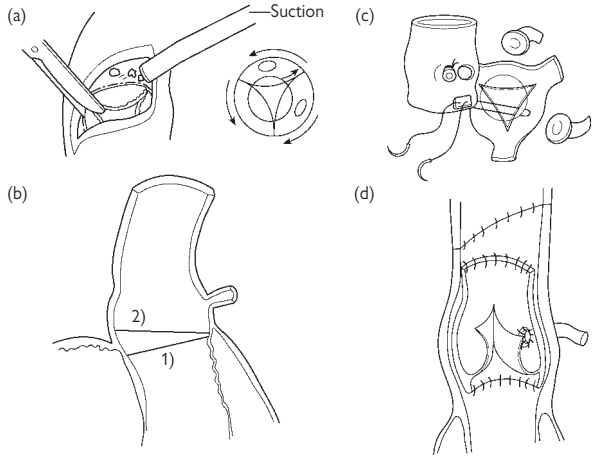


Fig. 8.5 (a) Excising the valve. (b) Aorta showing the relative position of 1) suprannular insertion and 2) subcoronary. (c) Cylinder implantation. (d) Root replacement.

Aortic valve implantation

Prosthesis selection is dealt with on 📖 p370, sizing is described on 📖 p380. Suture techniques can be divided into interrupted and semicontinuous.

Semicontinuous suture technique

Semicontinuous techniques require a non-absorbable monofilament such as 2/0 Prolene. One technique is described here: it can be used to implant prosthetic valves and homografts. With the assistant holding the valve, position the valve above the annulus and if not marked, mark the rough position of the commissures on the valve sewing ring. Suture underneath to above on the valve with the needle mounted backhand, and from above to below on the annulus with the needle mounted forehand.

- Take a single-ended 2/0 Prolene mounted forehand with a rubber shod on the cut end. Place the rubber shod under the retractor at 9 o'clock.
- Take a bite of the annulus from above down: usual place for starting is at the commissure between the right and the left coronary cusp.
- Next with the needle mounted back hand take a firm bite of the valve sewing ring from below up.
- Progress round the annulus and the sewing ring in an anticlockwise direction, spacing bites 2–3mm apart, with the assistant following so that all sutures are under tension to avoid you looping sutures, and so that the annulus is opened up by the sutures.
- If the prosthesis is held so that the area that you are suturing is always uppermost, even though both suture lines are progressing anti-clockwise, it looks as if they are progressing in opposite directions: this is why some surgeons call this a *counter-current* technique.
- Bring the prosthesis down towards the valve to maximize suture.
- Each time the suture becomes too short take an additional single-ended, clip the needle you are sewing with to the end of the new suture, cut just the needle off taking care not to create a loose remnant.
- You can only do this when the suture is on the upper aspect of the sewing ring: the first stitch with the new suture is into the annulus from above down.
- Use a leaflet retractor to separate sutures for the last few bites.
- Parachute the prosthesis down into position, by pulling alternately on each of the rubber shods and running the prosthesis down the sutures.
- Cut the stay stitch securing the valve to the valve holder with a 15 blade on a long handle, and remove the holder.
- Even out the sutures, tighten each in turn with a nerve hook checking carefully for loops, check that the valve mechanism opens freely, check that both coronary ostia are unimpeded, and then cut the sutures so that you can tie them comfortably.
- Place 8–9 throws so that the knot forms away from the valve, cutting the ends short so they do not interfere with the mechanism.
- As this is a semicontinuous technique it is sensible to maintain traction on the sutures either side of those being tied, but great care must be taken not to tear the proximal end of the aortotomy at this point by over-vigorous assistance.

Interrupted non-everting mattress suture

Interrupted sutures may be simple or horizontal mattress sutures (which may be everting or non-everting), with or without pledgets. Alternating color braided non-absorbable sutures such as 3/0 Ethibond are usually used. This technique is suitable for prostheses and homografts: horizontal mattress sutures tend to be used with prosthetic valves (everting in large annulus, non-everting in smaller, with pledgets where the annulus is very calcified or friable), and simple interrupted monofilament sutures for homografts. Either all the annular sutures are placed before putting any sutures through the prosthesis sewing ring (simpler), or each suture can be placed through the annulus and directly into the valve sewing ring.

- Take a double-ended 3/0 Ethibond with both needles mounted forehand hook, and a pledget mounted between them, and take a bite from below up on the annulus starting at the L/NCC commissure, repeating this 3mm away with the second needle and clipping both ends of the suture under tension to the drapes or valve holders, after making sure the pledget seats correctly and suture 'runs' freely.
- Repeat with the first stitch of the second suture 1mm away from the second stitch of the first suture, proceeding around the annulus and the valve sewing ring in an anti-clockwise direction. Usually easier to use a backhand from the middle of the RCC to the middle of the LCC from the right side of the table, or take forehand from the left side.
- Most surgeons alternate two colours of sutures and clip both ends of each suture together, some put a single throw on the sutures before giving them to the assistant, and others prefer to rely on the spacing on the valve sewing ring to tell which suture ends are the correct pairs.
- With the assistant holding the valve mounted on a holder, position the valve above the annulus. Note the number of sutures. Both ends of the double-ended sutures are placed through the valve sewing ring from below up. Securing the pledget above the annulus is what *everts* the annulus: this is bulkier than the semicontinuous technique and non-everting techniques. In the smaller annulus pledgets should be placed below the annulus, or not used.
- Make a rough estimate of how many sutures you'll need per commissure, to help you plan the spacing on the valve sewing ring.
- The assistant holds the sutures in their right hand once placed through the sewing ring to keep them under tension, and provide exposure.
- If for any reason a suture is cut before it has been placed through the valve sewing ring, a French-eye needle or standard green needle can be used to threat the loose end through the valve ring.
- When the suture ring is complete, gather the sutures together between two or three clips, and cut the needles off together, taking care not to lose any needles or to create any loose suture fragments.
- Wet the sutures with saline.
- Holding the sutures taut, parachute the valve into position, and complete implantation as described for semicontinuous technique: with braided sutures five throws is sufficient for each knot, and check each suture 'runs' before tying to avoid tying to the wrong suture, and leaving loose loops of suture below the annulus.

Stentless aortic replacement

Stentless aortic prostheses can be implanted in the subcoronary position which is described here, as a cylinder insertion or, in the case of the Freestyle, implanted in the same way as an aortic homograft root replacement (📖 p390).

Subcoronary technique

The Medtronic Freestyle® needs to be trimmed before it can be implanted using this technique, whereas the 3f is ready for use. The right and the left coronary sinus are removed, the non-coronary sinus is left intact (Fig. 8.6). If an oblique aototomy has been used, the non-coronary xenograft sinus helps to buttress the closure.

- Transverse aortotomy. Excise and size the valve in the usual manner.
- Valves are sized according to the outside diameter of the prosthesis, as opposed to homografts which are sized according to the inside diameter. It is therefore not necessary to downsize: oversizing the stentless valves actually increases the gradient across the valve because of extra tissue occupying the orifice.
- It is particularly important to get the orientation of the prosthesis relative to the aortic root correct, as the patient's coronary ostia will be obscured if the orientation is incorrect: coronary arteries in the pig are closer together than in the human.
- The valve can be inserted using either interrupted or a semi-continuous technique, with or without pledgets: as the sewing ring is less rigid it is important to check that suture tension is constant throughout and that the sewing ring is not distorted.
- More so than for stented and mechanical prosthesis, the horizontal sewing ring means that sutures should remain in a flat plane on the aortic annulus, to avoid distorting the flexible sewing ring (Fig. 8.6c).
- The porcine aorta is attached to the patient's aorta using semicontinuous non-absorbable monofilament such as 3/0 Prolene (Fig. 8.6d) (3x single interrupted mattress sutures, one at each post, in the case of the 3f stentless valve).

Aortic root replacement technique

The additional dissection required to isolate the patient's coronary ostia and create coronary buttons, and the additional suture lines required, mean that this is a longer and more technically demanding operation than subcoronary implantation. When the aortic root is dilated or distorted, it is preferable to insert the Freestyle® as a root replacement. The coronary ostia are closer together in the pig than they are in the human: when the left coronary ostia of xenograft and patient are aligned, the right coronary ostia in the xenograft is usually to the right of the patient's. It is usually necessary to extend the hole in the xenograft to match (Fig. 8.6), otherwise implantation follows the steps described for homograft root replacement on 📖 p386.

Cylinder implantation technique

This is described on 📖 p387.

Indications for stentless AVR

- Subcoronary: small aortic root, surgeon preference.
- Root replacement: see 📖 p390.

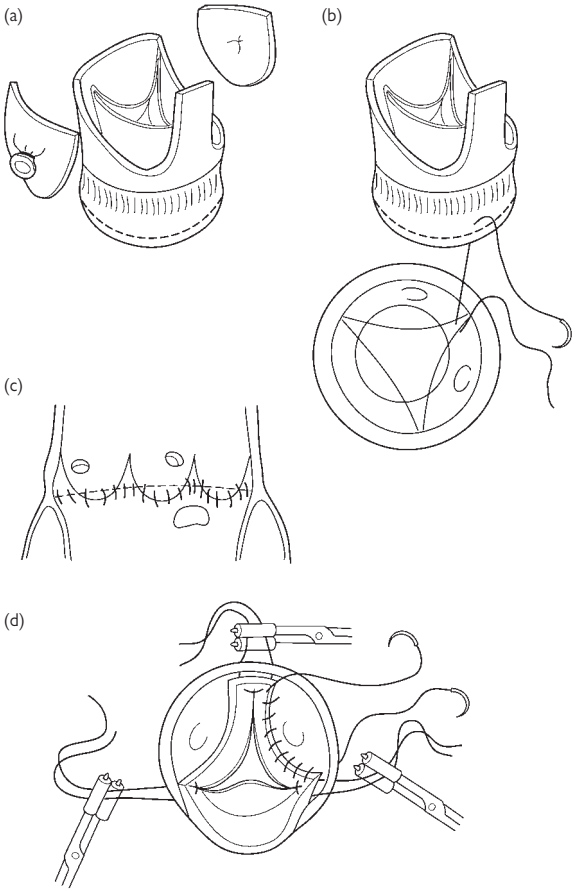


Fig. 8.6 (a) Preparing the Freestyle prosthesis. (b) Implantation of the stentless xenograft; (c) Placement of stitches on the aortic annulus; (d) Attaching the porcine aorta to the patient's aorta, with marker stitches and semicontinuous technique.

Homograft aortic replacement

Aortic homograft (also known as *allograft*), can be inserted in the same way as a Freestyle prosthesis as a subcoronary aortic valve replacement (📖 p384), as an inclusion cylinder (📖 p387), or as a root replacement (📖 p390). To preserve the architecture of the pliable homograft root, it is most commonly inserted as a root replacement. There are a few differences between Freestyle and homograft insertion (see Box 8.3).

Homograft preparation

This is described on 📖 p368. The homograft can be trimmed as described on 📖 p384 for subcoronary implantation, or prepared for cylinder implantation or root replacement (Fig. 8.6).

Homograft root replacement

- The aorta is transected and coronary buttons dissected free (Fig. 8.6).
- The aortic valve is excised, and the valve sized: the homograft should not be more than 3mm smaller than the patient's annulus.
- The homograft is used in its natural orientation.
- Implantation technique is an interrupted non-everting horizontal mattress, with non-absorbable monofilament sutures such as 3/0 Prolene.
- The coronary buttons are attached carefully to the site of the excised ostia of the homograft using continuous 5/0 Prolene. Great care must be taken to avoid damaging or kinking the coronary arteries (📖 p390).
- The root replacement is completed by end-to-end anastomosis of the homograft to the native aorta, usually with the clamp on: if the aorta is aneurysmal or heavily calcified a more distal anastomosis may be performed under circulatory arrest with the clamp off (📖 p435).

Box 8.3 Differences between Freestyle and homograft

- The size of homograft must be checked preoperatively as not all sizes will be kept in stock: usually Freestyle is kept in all sizes.
- Homografts are sized according to the internal diameter: Freestyle valves are sized according to the external diameter.
- The homograft must be carefully prepared before insertion (Fig. 8.6), there is a risk of damaging the aorta and the valve leaflets.
- The homograft is much more pliable than the Freestyle, and even greater care has to be taken not to distort its architecture.
- Because the tissues are so pliable it is sometimes easier to invert the homograft when implanting it in the subcoronary position and sew it *in situ* (Fig. 8.6), rather than parachuting it into position.
- The polyester fabric provides a firmer sewing ring than the homograft tissue, which can be very friable.
- There is less mismatch between the coronary ostia of the patient and the prosthesis in homograft AVR compared to Freestyle AVR.

Cylinder technique

- The homograft aorta is prepared by excising larger buttons of sinus aorta than in the root replacement technique. It is implanted in a similar way to the root replacement, but the native aorta is preserved so that the aortotomy can be closed directly (Fig. 8.6). This technique is used less commonly as there is a greater tendency to distort the valve geometry than with a complete root replacement technique.

Indications for homograft root replacement

- Infective endocarditis with root abscess requiring root replacement: not so much because of increased freedom from reinfection (which has not been shown in recent series), but because homograft is slightly easier to sew into friable tissue (particularly reop) than bio-Bentall, or Freestyle.
- Similar rationale for reop aortic root replacement—need to mobilize coronary buttons less extensively as homograft pliable.
- Surgeon preference is less common now as there is some data to suggest poorer durability of homografts compared to Freestyle root replacement.

The Ross procedure

Using the patient's own pulmonary valve to replace the aortic valve (pulmonary autograft) is known as the Ross procedure, after Donald Ross, the British surgeon who pioneered its use. The pulmonary valve is replaced with a pulmonary homograft. Rationale is discussed on 📖 p369.

Set-up and aortotomy

Some surgeons prefer no PA catheter. Aorto-bicaval cannulation (the RV is opened during the procedure so use of a two-stage cannula, may: (a) entrain air, and (b) result in poor venous drainage and a flooded field). Mark the pulmonary valve commissures before going on bypass.

Excising the PA root and preparing the aorta

- Assess the pulmonary valve by TEE, and then by direct visualization through small arteriotomy: large or multiple fenestrations, or bileaflet valves are not suitable for use as autograft.
- After clamping separate the PA from the aorta up to the bifurcation, and proximally as far as the area where they are fused adjacent to the right–left aortic commissure, just onto the RVOT muscle.
- Transverse aortotomy 1–2cm above commissures, transecting aorta if suitable for root replacement, and excising valve in standard way (📖 p380).
- Size valve using standard sizers, and place six stay sutures: three above each commissure, two above R&L coronary ostia, and one deep in NCC.
- Divide PA 1–2cm above the sinuses: there is no advantage to using the entire length of the PA as this has a tendency to dilate in the aortic position, and getting good exposure for a very distal pulmonary artery to homograft anastomosis is quite challenging.
- Divide fat between back of PA and LMS carefully staying close to PA.
- Open RVOT just below PA cusps, extending incision medially and laterally staying well away from cusps, keeping 3–5mm of muscle cuff, and staying in the natural plane between right and left ventricular septum posteriorly to avoid damaging the first septal artery which is located in the LV septum, and easily damaged during this dissection, compromising septal function and resulting in much of the morbidity associated with the procedure.
- The pulmonary autograft is sized and trimmed ready to be implanted.
- The aorta is transected and large coronary buttons mobilized (sharing the aortic wall equally between the buttons and annulus, and preserving pillars of native aorta).

Implantation of the pulmonary autograft

- Ideally <2mm difference between aorta and the pulmonary autograft.
- The pulmonary autograft is implanted in a similar fashion to the homograft root replacement described on 📖 p386 using semicontinuous 4/0 Prolene on RB1 and SH1 needles.
- Supporting the aortic annulus with a strip of Teflon felt or pericardium cut 1 size larger than the internal diameter helps prevent dilatation.
- If diameter of the aorta is larger than the PA this can be addressed either by bevelling the pulmonary trunk or by using interposition grafts.

Indications for Ross procedure

- Patient <21 years of age requiring AVR.
- Controversial: adults who do not wish to take warfarin in whom Ross can be performed with mortality close to that of isolated AVR.

Implantation of the pulmonary homograft

The PA can be anastomosed before the aorta as exposure is easier, or afterwards to reduce ischemic time: both are relatively straightforward full-thickness continuous 4/0 Prolene, except for the area around the infundibular septum where sutures must be partial thickness to avoid damaging the first septal branch of the LAD. Both suture lines can be buttressed with pericardial strips—bleeding is pressure dependent so ensure the anesthesiologist appreciates the need to keep systolics <100mmHg at all times to avoid major hemorrhage.

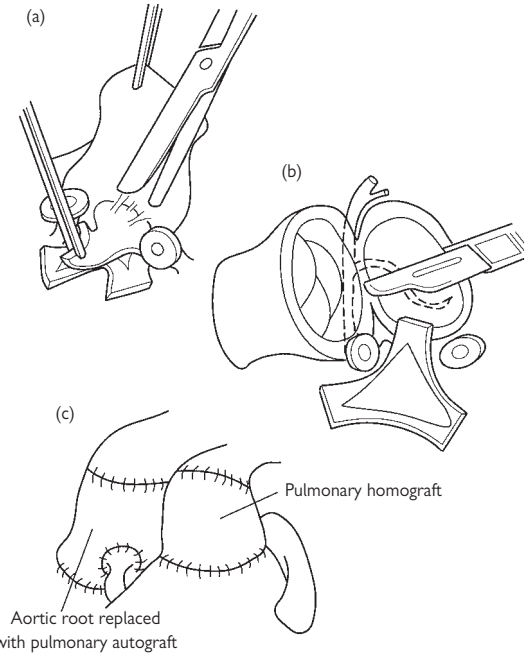


Fig. 8.7 (a) Preparation of the aortic root and dissection of the infundibular septum. (b) Preserving the first septal branch of the LAD. (c) Completed Ross.

Aortic root replacement

Key facts

- Can be done using homograft, Freestyle porcine valve, pulmonary autograft (Ross), or most valved conduit (Bentall or bio-Bentall).
- Performed for annuloaortic ectasia/aneurysm where native valve cannot be preserved (spared), annular abscess, small aortic root (📖 p392).

Indications for root replacement

- Annuloaortic ectasia/aneurysm aortic root where valve cannot be preserved. Size cut-offs listed in more detail on 📖 p443.
  - Ascending 5.5cm asymptomatic incidental.
  - Ascending 5cm symptomatic.
  - Ascending 4.5cm Marfan (or symptomatic), bicuspid aortic valve.
- Infective endocarditis of the aortic root—annular abscess.
- Acute or chronic aortic dissection with aortic regurgitation (📖 p432).
- Small aortic root, to allow larger prosthesis than otherwise possible.

Set-up

The arterial cannula may be placed in the distal ascending aorta, axillary or femoral artery (📖 p433)—check! If possible separate the pulmonary trunk from the back of the aorta: when the aorta is aneurysmal this is challenging. CPB is established using a single two-stage cannula, and the patient cooled to 28°C (18°C if the distal aortic anastomosis is to be performed with the clamp off). The heart can be vented in any of the ways listed on 📖 p378. The aorta is cross-clamped and cardioplegia may be given antegradely if there is minimal AI. If there is moderate to severe AI other options are to arrest with retrograde, or open the aorta to give blood directly into the coronary ostia (📖 p378).

The Bentall procedure

Homograft aortic root replacement is described on 📖 p386, and aortic root replacement with pulmonary autograph is outlined on 📖 p388. A more commonly used alternative is composite valve conduit. This consists of a mechanical aortic valve sewn into the base of a tubular Dacron graft (Bentall procedure), or in the case of a bio-Bentall any stented prosthesis may be sewn into the bottom of a Dacron conduit at least one size larger, using a running 4/0 Prolene suture).

- Place six traction stitches: three above each commissure, two well above each coronary ostia and one deep in the non-coronary sinus to expose the root (Fig. 8.7), and excise the aortic valve.
- Mobilize the coronary arteries retaining a generous button of sinus aorta, so that they can be implanted into the Dacron graft later without kinking or stretching, but sharing the aorta at the nadir of each button equally with the annulus so that you have something to put your annular sutures into. Remove the remaining sinus aorta (Fig. 8.7).
- Size the annulus, and the approximate length of ascending aorta to be replaced and select an appropriate graft (usually size 26 or 28).

- Implant the composite graft using either of the techniques described for simple AVR on 📖 pp382–383. If tissues are friable, use interrupted pledgetted sutures (Fig. 8.7). In a bio-Bentall, annular sutures should be passed through the sewing ring of the conduit and the bottom 2mm of the conduit: it is this suture line that makes the join between graft and prosthesis hemostatic, not the 4/0 running Prolene.
- Next assess where the left coronary button sits, and create a 4mm hole in the graft with microcautery to which the left coronary button can be anastomosed with a 6/0 Prolene continuous suture line, which can be buttressed with a felt strip (starting halfway up the left side of the anastomosis forehand on the graft inside to out, outside to in on button, working past the heel and halfway up the right side before swapping ends of the suture). This is checked for hemostasis by applying a clamp distally and giving cardioplegia directly into the graft.
- The position of the right coronary button should be determined with the perfusionist filling the heart to avoid the proximal RCA kinking (suture with 6/0 Prolene starting halfway up the left side of the anastomosis forehand on the button inside to out, outside to in on the graft, working past the heel and halfway up the right side before swapping ends of the suture).
- In large aneurysms the coronary ostia may displaced so far laterally that an interpositional or Cabrol graft is required (📖 p445).
- Next shorten the distal end of the tube graft to match the distal aorta. Anastomose with continuous 4/0 Prolene (starting halfway up the left side of the anastomosis forehand on proximal inside to out, outside to in on distal, working past the heel and halfway up the right side before swapping ends of the suture). Tighten the suture line with nerve hooks, especially posteriorly where if bleeding occurs it will be most problematic to rectify. For the same reason, if there is a size mismatch cheat anteriorly not posteriorly.

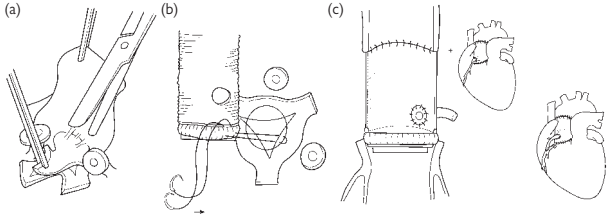


Fig. 8.8 (a) Exposure of the aortic root and creating coronary buttons. (b) Implanting the composite conduit. (c) Finished repair.

Small aortic root

Key facts

- The post-op gradient across an AVR depends on the valve size and CO: it is higher in small valves, large patients, tachycardia, hyperdynamic LV (hence β-blockers used to treat it).
- Post-op gradients are common, and it's probably sensible to minimize gradient by not implanting an AVR 'too small' for the patient's size.
- How small is 'too small' is quantified by *patient prosthesis mismatch (PPM)*, defined as indexed effective orifice area (IEOA)<0.85cm^2/m^2:
  - There is, however, no strong data to show difference in survival or symptoms according to whether patients have PPM.
  - This is partly because PPM tends to occur in patients with other risk factors for worse outcomes (females, obese, older patients).
- There are several options to avoid a high transvalvular gradient after AVR, in a patient at risk of this because of a small aortic root (Box 8.4).

Patient prosthesis mismatch

In vivo valve orifice area and body size are used to calculate IEOA: <0.8cm^2/m^2 results in a rapid rise in mean transvalvular pressure gradient during exercise, reflected by reduced LV mass regression postoperatively. No studies show strong evidence of worse survival or symptoms due to PPM. The consensus is that a 19mm valve should not be implanted in a large, active patient because of the large intrinsic gradient. Inserting the largest possible prosthesis may cause problematic annular erosion acutely or over time. Alternatives are listed in the Box 8.4: it is a question of balancing the theoretical disadvantages of moderate AS against the disadvantages of the surgical options.

Aortic root enlargement

The oblique aortotomy described on 📖 p379 has the important advantage that it can be easily extended to allow aortic root enlargement, and can be easily closed with a patch enlargement. Root enlargements are either posterior (used mainly in adults) or anterior (used in children).

The Nicks procedure

The aortotomy incision is continued posteriorly through the NCC, onto the anterior leaflet of the mitral valve, and the roof of the LA is dissected away without opening it. Dividing the fibrous attachment of the aortic

Box 8.4 Options for small aortic root (simplest first)

Plan ahead: annular diameter <20mm on echo is likely to take only 19mm valve—will this be adequate? If patient very active or high BSA, or diastolic dysfunction, or low-gradient low-EF AS—probably not.

- Implant mechanical valve (best hemodynamics for given size: but trading lifetime anticoagulation for a reduction in gradient).
- Suprannular stented valve (allows to upsize by one size).
- Stentless valve (slightly better hemodynamics than stented).
- Nicks root enlargement.
- Manougian root enlargement.
- Root replacement.
- Ross–Konno (enlargement via RVOT used in children only).

annulus in this way causes the edges of the aortotomy to open widely as there is little fibrous support. A 3–4cm wide teardrop-shaped patch of either pericardium or Dacron is shaped to fit the defect. Starting at the midpoint of the bottom of the incision the patch is sutured to the aorta, with or without Teflon pledgets (Fig. 8.8a). This must be done particularly carefully as this area is virtually impossible to access once the cross-clamp is removed. The valve is inserted in the usual manner, sewing it to the aortic annulus and the patch.

The Manougian procedure

This involves extending the aortotomy into the posterior commissure, between LCC and NCC into the interleaflet triangle, and onto the anterior leaflet of the mitral valve. The loose tissues of the triangle separate readily, and it is often but not always possible to enlarge the annulus sufficiently without having to divide the anterior leaflet of the mitral valve or enter the LA.

The Ross–Konno procedure

This is the Ross procedure (📖 p388) with anterior enlargement of the root.

Root enlargement using a homograft: Root replacement always allows a larger valve to be implanted: if even greater enlargement is needed (e.g., in reoperative surgery) the anterior mitral annulus may be divided. A homograft, complete with anterior mitral valve leaflet if necessary, is implanted in the subcoronary position, with closure of the roof of the LA with a pericardial patch, and pledget strips to buttress unsupported aorta. The NCC of the homograft is used to close the aortotomy and widen the aortic root.

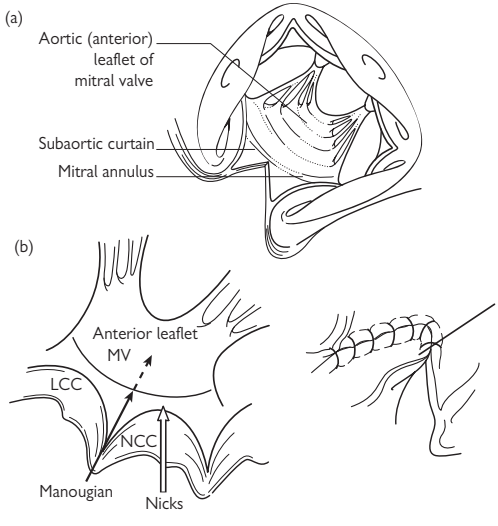


Fig. 8.9 (a) Extending the oblique aortotomy for root enlargement (Nicks procedure). (b) The Nicks and Manougian posterior root enlargement compared.

Valve-sparing procedures

There are a large number of procedures described for patients with morphologically normal aortic valves, but with AI caused by pathology affecting the sinotubular junction (STJ) or the aortic annulus (AA). The differences between and indications for the Yacoub (remodelling) and David (reimplantation) valve sparing procedures are described on 📖 pp446–447. The normal sinotubular junction, and annulus are described on 📖 p53.

Indications for valve-sparing procedures

- Normal AA and STJ: Commissuroplasty, leaflet plication.
- Normal AA and ↑STJ: interpostion graft if sinuses do not need replacing (📖 p443).
- ↑AA and ↑STJ: Yacoub or David procedure (📖 pp446–447) (Fig. 9.6c).

Normal AA and normal STJ

AI may be due to leaflet abnormalities (perforation, fenestration or prolapse) which can occasionally be satisfactorily addressed by primary repair and/or leaflet plication using 6/0 Gore-Tex or 7/0 prolene. Aortic commissuroplasty involves reducing the intercommissural triangle distance with horizontal pledgted mattress sutures. When AI is due to distortion of a sinus by Type A dissection (Fig. 8.10a, valve sparing options include sealing the dissected sinus with sutures or glue and replacing the ascending aorta with an interposition graft (Fig. 8.10e, and 📖 p444)—this is the preferred option; or a valve sparing root replacement (Fig. 8.10b) with coronary reimplantation which is technically extremely challenging in this context.

Normal AA and dilated STJ

This occurs in patients with aortic ectasia, which may be caused by connective tissue disorders such as Marfan, chronic hypertension, post-stenotic dilatation, and type A dissection. A simple interposition style graft can be placed, with or without a band of PTFE for additional support at the level of the STJ (Fig. 8.10e). It is not necessary to reimplant the coronaries.

Both AA and STJ enlarged

This is a feature of annuloaortic ectasia which is caused by connective tissue disorders such as Marfan. A number of modifications to procedures originally separately described by Magdi Yacoub and Tirone David exist. The main differences (see also 📖 pp446–447) between the various modifications are:

- How the size of aortic graft required is calculated.
- Whether an aortic annuloplasty is performed.
- Whether the aortic annulus is fixed circumferentially.
- Whether an inclusion cylinder technique is used.
- Whether and how pseudosinuses are created in the graft.

Checking competence

This can be assessed prior to completing the distal aortic anastomosis by holding the graft in the approximate anatomical position and filling it with saline, which should remain in the root. Formal assessment occurs with intraoperative TEE once the patient is off bypass.

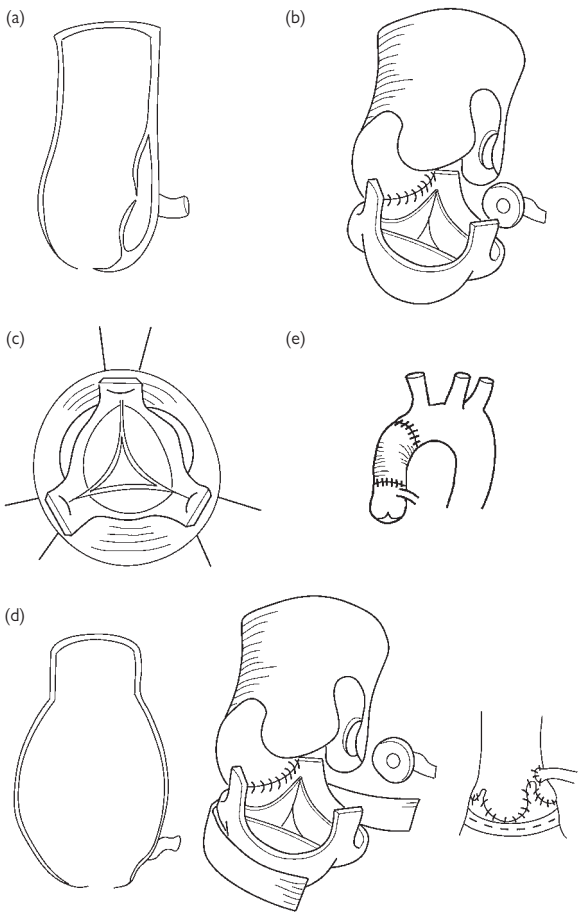


Fig. 8.10 (a) Cross-section of aortic dissection. (b) Creating pseudosinuses. (c) Resuspending the aortic valve inside the Dacron graft. (d) Valve-sparing surgery for true aortoannular ectasia with PTFE support of annulus. (e) Simple interposition graft.

Results of aortic valve surgery

Key facts

- Operative mortality of isolated AVR in patient <75 years ~1%.
- Stroke ~1%, re-exploration ~1%, complete heart block ~1%.
- Bioprosthesis failure (📖 p366): 10–20% by 12 years, 1/3 by 20 years.
- Mechanical valve reoperation (pannus, SBE) ~10–15% by 20 years.

Early mortality

The following data is from 409 904 isolated valve procedures in the USA national database (1994–2003).[1,2] Unadjusted mortality includes all comers—reops, low EF, emergency, elderly, etc.:

- Single AVR (including CABG) unadjusted mortality 5.7%.
- Root replacement: unadjusted mortality 11.1%.
- Biggest risk factors for mortality include emergency (odds ratio (OR) 2.1), age >70 (OR 1.9), multiple valve (OR 2.0), CAD (OR 1.6) aortic root (OR 2.8), endocarditis (OR 1.6).
- Results improved between 1997–2006 despite sicker patient profile: for AVR predicted mortality increased from 2.8% to 3.2% whereas observed mortality fell from 3.4% to 2.6%, and stroke rates decreased from 1.5% to 1.3%.

Most early deaths are related to cardiogenic shock, CVA, hemorrhage. The higher mortality in patients undergoing bioprosthetic AVR is related to their higher preoperative comorbidity, including greater age. Euroscore weights the preoperative risk factors for early mortality (📖 p358).

Late mortality

Overall survival (including hospital deaths) after AVR is approximately:

- 75% at 5 years.
- 60% at 10 years.
- 40% at 15 years.

Short-term survival is dictated primarily by perioperative mortality. After about 6 months after operation death rate steadies at ~2% pa. This begins to rise steadily after 5 years (late-term survival). Patients with bioprosthetic valves have significantly worse survival curves than patients with mechanical valves: as with early mortality, this reflects the greater age and comorbidity of this group. Other risk factors for late mortality include:

- ↑age at operation (30-day mortality 1% age 40, 8% age 75) (📖 p358).
- Afro-Caribbean origin, female sex.
- Higher NYHA class, LV enlargement, grade of AR, angina, and AF.
- Number of previous AVRs, coexisting CAD, aneurysm, or LV structural abnormality.
- The operative myocardial ischemic time.

Complications of aortic valve replacement

The complications of cardiac surgery in general: exploration for bleeding/tamponade (1%), perioperative MI (1%), deep sternal wound infection (1%), prolonged mechanical ventilation (2%), and organ dysfunction (5%)

are discussed in more detail on 📖 pp253–288. Complications particularly associated with AVR include:

CHB: permanent pacemakers are inserted for permanent CHB in about 1% of AVR patients. CHB results from trauma to the bundle of His after extensive debridement of calcium from the RCC/NCC junction (📖 p380). Worsens day 1–3, settles as edema improves.

CVA: the overall risk of CVA is about 2% for isolated AVR, increasing to 3% in combined AVR + CABG. CVA in AVR shares the same risk factors as in other cardiac surgery (📖 p278). Removal of all calcific debris from the operative field and thorough de-airing help prevent CVA.

Prosthetic endocarditis: mechanical valves are at slightly higher risk of endocarditis up to 3 months after surgery; the risks even out by 5 years. Homografts once thought lower risk, not supported by data. The cumulative risk of prosthetic valve endocarditis is about 1% at 1 year and 3% at 5 years post AVR. Early endocarditis (within 3 months of surgery) is usually hospital acquired: organisms are commonly staphylococci and Gram −ve cocci. Late prosthetic endocarditis usually results from a transient streptococcal bacteremia. The mortality is up to 60%.

Thromboembolism: the rate of valve thromboembolism is roughly 1–2% per year (📖 p267): the rate in adequately anticoagulated mechanical valves is similar to that in prosthetic valves. *Valve thrombosis* rare emergency: cardiovascular collapse, venous congestion and systemic embolization normally acute, but may be insidious. The management is iv heparin. If the thrombus is >5mm thrombolysis or reop tissue AVR is indicated.

Paraprosthetic leak: major paraprosthetic leak is uncommon (📖 p266). It is caused most commonly by prosthetic endocarditis (late), and less commonly by technical errors in insertion (early). Patients present with hemolytic anemia, and if severe, CCF. The treatment is surgery. If the area of dehiscence is small and there is no evidence of sepsis, one or two interrupted sutures may suffice. Otherwise redo AVR is indicated.

Prosthesis failure (📖 p366)***:*** failure of mechanical valves is almost exclusive to strut failure of a particular Bjork–Shiley tilting disk valve, withdrawn in 1986. Bioprosthetic AVR failure is <10% at 10 years in the over 70s, but closer to 30% in the 40–50-year age group (📖 p366). Degenerative failure in cryopreserved homograft AVR is between 5–10% at 10 years.

Reoperation for any reason: the risk of reoperation is highest in the first 6 months post AVR due to prosthetic endocarditis and paravalvular leak. It falls to about 1% a year where it remains for mechanical valves. The risk also falls initially in bioprosthetic valves, but begins to rise rapidly after 10–12 years due to structural degeneration.

Improvement in symptoms and LV function

90% of patients are NYHA class I or II, 5–10 years following AVR. Completely normal LV mass is rarely achieved, but where LVEDP was low preoperatively, LV mass returns to near normal values. Where LVEDP was raised preoperatively this does not happen.

References

1. Rankin JS. Amiodarone and cardiac surgery. *J Thorac Cardiovasc Surg* 2005; **131**:547–57.
2. Brown JM *et al.* Isolated aortic valve replacement in North America comprising 108,687 patients in 10 years. *J Thorac Cardiovasc Surg* 2009; **137**: 82–90.

Pathophysiology of mitral stenosis

Key facts

- Commonest valve lesion in developing countries.
- Usually caused by rheumatic valve disease: results in commissural fusion, leaflet thickening and calcification, and chordal fusion.
- Well tolerated with long asymptomatic interval.

Morphology

The commonest cause of MS is rheumatic fever: the disease process causes leaflet thickening and calcification, commissural and chordal fusion. Isolated MS occurs in 40% of all patients presenting with rheumatic heart disease, and 60% of patients with MS give a history of rheumatic fever. The ratio of female to male patients with this pathology is 2:1. Congenital malformation of the mitral valve occurs rarely (📖 p638).

Structural and functional changes

The cross-sectional area of the normal mitral valve is 4.0–5.0cm^2. Narrowing of this area to <2.5cm^2 means that a pressure gradient must be generated across the mitral valve in diastole in order to expel blood from the LA into the LV. This results in elevation of LA and pulmonary venous pressures. LA enlargement, AF, and thrombus formation result from chronic LAP overload. Pulmonary hypertension results from passive transmission ↑LAP, pulmonary vasoconstriction and irreversible pulmonary vasculature remodelling. Pulmonary edema occurs when pulmonary venous pressure is greater than plasma oncotic pressure. At rest, MS is usually asymptomatic, but any increase in transmitral flow rates (e.g., due to hyperdynamic circulation) or reduction in the diastolic filling period, e.g., tachycardia will result in an increase in the pressure gradient, resulting in dyspnea. Severe MS may have a low gradient in patients with a low CO.

Symptoms

The first symptoms of dyspnea in these patients are usually precipitated by exercise, sexual intercourse, emotional stress, infection, pregnancy, or rapid AF, because in the increase in the pressure gradient across the mitral valve described earlier. Recurrent chest infections are characteristic. Hemoptysis is caused by chest infections, pulmonary infarction, acute pulmonary edema, and rupture of small pulmonary vessels. Systemic embolism is frequent, particularly in AF.

Signs

- Decreased amplitude pulse, AF.
- 'Tapping' apex beat.
- Left parasternal heave due to RV hypertrophy.
- Thrill due to paplpable opening snap.
- Loud first heart sound, an opening snap loudest at lower left sternal edge.
- Presystolic murmur, delayed mid-diastolic murmur.

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Pathophysiology of mitral regurgitation

Key facts

- Commonest valve lesion with incidence around 2%.
- Describe using *Carpentier's pathophysiological triad* (see Table 8.3):
  - *Leaflet dysfunction*: leaflet motion either I normal, II excessive, or III restricted (IIIa restricted opening, IIIb restricted closure).
  - *Lesion*: problem with MV apparatus, e.g., chordal elongation or rupture, leaflet perforation, annular dilatation, papillary muscle rupture.
  - *Etiology*: degenerative, ischemic, rheumatic, endocarditis, DCM.
- Mitral *prolapse* is systolic billowing of leaflets into the LA ± MR.
- MR causes dyspnea, CHF, AF, stroke, pulmonary hypertension.

Morphology

- The mitral valve consists of annulus, leaflets, chordae tendineae; papillary muscles; and the LV wall (Fig. 2.5). Disease involving these parts results in reduced leaflet coaptation leading to MR, described using Carpentier's pathophysiological triad (Table 8.3) and *segmental valve nomenclature* (📖 p45).

Etiology

- *Degenerative disease* is commonest cause in West (incidence 2%):
  - Causes prolapse (type II MR) due to chordal elongation or rupture.
  - Histology shows *myxomatous degeneration.*
  - Spectrum from single-segment prolapse in small valve (*fibroelastic deficiency*) to multisegment prolapse in giant valve (*Barlow's*).
- *Ischemic MR* is usually due to chronic LV dysfunction and posterior displacement of papillary muscle (type IIIb) with normal leaflets; papillary muscle rupture can cause acute type II MR after MI.
- *Endocarditis* causes type I MR (leaflet perforation).
- *Dilated cardiomyopathy* causes type I MR (annular dilatation).

Table 8.3 Carpentier's pathophysiological triad

	Leaflet dysfunction	Lesions	Etiology
CARPENTIER'S CLASSIFCATION OF MR AC	Type I	Leaflet perforation	Endocarditis
		Annular dilatation	Dilated cardiomyopathy, ischemic cardiomyopathy
	Type II	Chordal elongation/rupture, papillary muscle elongation	Degenerative disease (FED or Barlow's)
		Papillary muscle rupture	Ischemic cardiomyopathy
	Type IIIa	Commissural fusion, leaflet calcification, chordal fusion	Rheumatic disease
	Type IIIb	Papillary muscle displacement, LV dilatation	Dilated cardiomyopathy, ischemic cardiomyopathy

- *Rheumatic disease* results in type IIIa MR with restricted leaflet opening due to calcification and commissural fusion.
- *Systolic anterior motion* (📖 p191) results in MR + LVOTO.
- *Congenital,* e.g., cleft mitral valve, *connective tissue disease,* e.g. Marfan.

Structural and functional changes in chronic MR

Whatever the mechanism resulting in reduced leaflet coaptation, the pathophysiological changes progress through one or more of three stages: acute MR, chronic compensated MR, and chronic decompensated MR.

- *Acute MR* results in *LV volume overload* at end diastole, i.e., ↑*preload*, and ↓*afterload* due to the regurgitant pathway into the LA.
- The combination of ↑preload and ↓afterload means that a larger volume of blood is ejected from the LV.
- Because a large volume of ejected blood enters the LA rather than the aorta, forward stroke volume, and hence *cardiac output decreases*.
- The increased blood volume in the LA raises the pressure from a normal LAP of 10mmHg to up to 25mmHg.
- ↑preload leads to eccentric hypertrophy and *LV dilatation*, increasing total stroke volume, as well as forward stroke volume which returns to near normal levels: *chronic compensated MR.*
- *LA enlargement* accommodates volume overload there at lower filling pressures (15mmHg), but predisposes to *AF* and *mural thrombi.*
- *Chronic decompensated MR* exists when systolic dysfunction prevents effective ventricular contraction: stroke volume and cardiac output are reduced as blood flows into the low-resistance regurgitant pathway, and left atrial and *pulmonary pressures rise* as a result.
- Untreated decompensated MR rapidly progresses to pulmonary edema and congestive cardiac failure.
- Symptoms, LV dysfunction, LA dilatation, AF, and pulmonary hypertension are all associated with decreased long-term survival.

Symptoms and signs in chronic MR

The interval from diagnosis of MR to the onset of symptoms varies by aetiology (FED months to years, Barlow's decades). LV EF remains normal despite even severe LV dysfunction because 50% of stroke volume ejects into low-pressure LA. While most deaths are related to heart failure, the incidence of sudden death suggests ventricular arrhythmias are an important feature of the disease process. Signs include AF, prominent sustained apex beat, systolic thrill, right parasternal heave reflecting RVH due to PAH, quiet first heart sound, third heart sound audible, loud pansystolic murmur loudest at apex radiating to axilla.

Acute MR

Acute severe MR may be caused by chordal rupture, ischemic heart disease, or infective endocarditis, and it is an indication for urgent surgery. A sudden volume overload is imposed on the LV, increasing preload and resulting in a small increase in total stroke volume. Without the compensatory LA and LV dilatation seen in chronic severe MR, however, forward stroke volume is reduced, and CHF and cardiogenic shock may result. The patient with acute severe MR is always symptomatic.

Timing of mitral valve surgery

Key facts

- Severe MR reduces normal life expectancy: repair is the only treatment that can restore this to normal. Mitral replacement is associated with worse early and late mortality and morbidity than repair (Fig. 8.11).
- Symptoms, AF, PAH, and LV dysfunction or dilatation are each associated with worse survival: so they are all indications for surgery in MR.
- Early surgery may be indicated in asymptomatic patients with MR and none of the listed risk factors *if repair is very likely (>90%).*
- MS is better tolerated than MR, is rarely amenable to repair, and may respond to balloon valvuloplasty so the threshold for surgery is higher.

Mitral regurgitation

The prognosis of chronic MR depends on both the etiology and severity of the lesion. In patients with severe symptomatic chronic MR of any etiology the prognosis is poor. The average mortality rate is ~5% per year, and may be as high as 70% at 8 years.

Patients with isolated chronic mild to moderate MR are unlikely to require surgery, and therapy is aimed at treating the symptoms and preventing complications. AHA guidelines recommend that these patients are followed up on a yearly basis, with TTE if there are clinical grounds for suspecting a deterioration in LV or mitral valve function.

In chronic ischemic MR (📖 p356) the valve itself is usually anatomically normal and the regurgitation is secondary to papillary muscle and LV dysfunction following infarction. The prognosis is poorer than in MR caused by non-ischemic pathology. CABG may improve LV function and reduce MR. If LV function is very severe (LVESD >55mm or LVEF <30%), and repair/chordal preservation unlikely, surgery may not be beneficial (📖 p357).

Acute mitral regurgitation

Acute severe MR is an indication for urgent surgery. There is no scope for compensatory changes: CO falls, LAPs increase, and pulmonary edema and cardiogenic shock ensue (📖 p401).

Mitral stenosis

It may be as long as 20–40 years between the occurrence of rheumatic fever and the onset of symptomatic MS. Once symptoms develop it may be up to 10 years before these become disabling. In the minimally symptomatic patient, 10-year survival is 80%, which drops to 10–15% once symptoms become limiting. Once there is severe pulmonary hypertension, mean survival drops to <3 years. The mortality of untreated MS is due to heart failure in 60–70%, systemic embolism in 20–30%, PE in 10%, and infection in 1–5%. Patients with a mitral valve area of <1.5cm^2 and NYHA functional class III–IV symptoms should be referred for surgery. There is good evidence that patients with NYHA functional class I–II symptoms and a mitral valve area <1.0cm^2, and severe PAH (systolic PAP >60–80mmHg) benefit from surgery.

Indications for mitral valve surgery

- Class 1: acute severe symptomatic MR, chronic severe MR *and* NYHA class II–IV *or* mild-moderate LV dysfunction. Also NYHA class III–IV moderate to severe MS if balloon valvotomy not possible.
- Class IIa: asymptomatic chronic severe MR with normal LV *if* repair >90% likely, *or* new AF, *or* PA systolic>50mmHg at rest (60mmHg exercise). Also NYHA III–IV+ severe LV dysfunction *if* primary leaflet dysfunction *and* repair likely. Also severe MS *if* PAP >60mmHg.
- Class IIb: chronic severe MR due to severe LV dysfunction despite optical medical therapy + biventricular pacing.

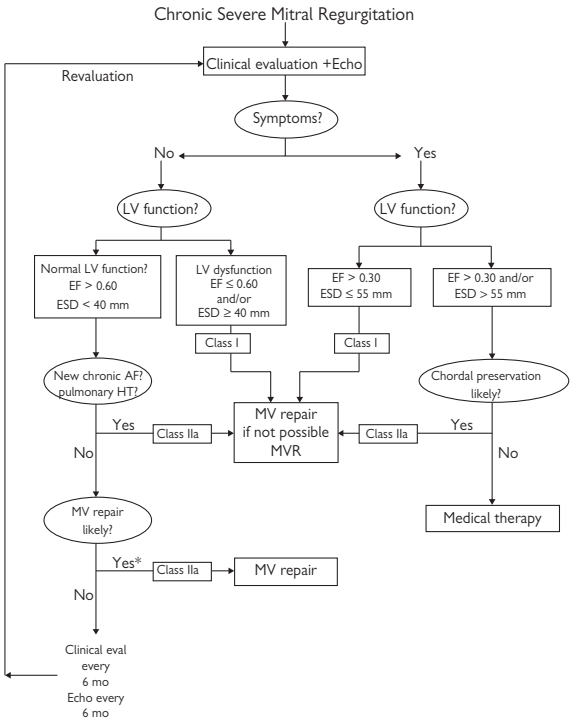


Fig. 8.11 American Heart Association/American College of Cardiologists indications for surgery in chronic severe MR. *Greater than 90% chance of repair. Reproduced from 2008 Focused Update Incorporated Into the ACC/AHA 2006 Guidelines for the Management of Patients With Valvular Heart Disease: A Report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*, 2008; **118**: e523–e661. Copyright © 2008, Wolters Kluwer Health.

Mitral valve exposure

Set-up

The commonest approach is via a median sternotomy. A right minithoracotomy via the 4th ICS is used for cosmetic reasons with more proximal cannulation of the ascending aorta and administration of cardioplegia or peripheral cannulation with endoclamping or fibrillatory arrest. Left thoracotomy is used occasionally. Only the right side of the pericardium should be suspended as this rotates the heart anticlockwise, lifting the LA up towards the surgeon. Incising the caval pericardial reflections inferiorly and superiorly further improves access. Rheumatic and endocarditic repair may require autologus pericardium (📖 p408).

Bypass

Standard aorti-bicaval bypass (CPB) at 32°C (📖 p110) (direct or via RA ± caval snares). Sternal mitral retractor, e.g., Cosgrove, McCarthy, or Wells.

Cardioplegia and venting

- Give anterograde cardioplegia to arrest the heart via a Medicut or Stockert cannula, retrograde useful during rest of case as no need to adjust retraction. Vent the LA directly, and aortic root during de-airing.

Approaches to the mitral valve

Seven or eight left atriotomy incisions: can be divided according to whether they involve a right atriotomy or not (Fig. 8.12):

- Right atriotomy: A, *Guiraudon's* (extended superior septal), B, *Dubost*, C, vertical transeptal, D, trans ASD.
- Left atriotomy only: E, paraseptal via *Sondegaard's groove*; F, classical paraseptal; G, superior septal; H, LA appendage; I, left thoracotomy.
- No atriotomy: J, transaortic; K, transventricular; (L, autotransplant).

Paraseptal left atriotomy: most commonly used incision. Retract the lateral RA wall so that the interatrial groove between the back of the RA and the roof of the LA known as Waterston's or Sondergaard's groove, can be developed. This simple maneuver improves exposure of the mitral valve. If the LA is enlarged dissection in this groove is less necessary as a larger left atriotomy is possible giving good access. In this case the incision starts more medially (Fig. 8.12, F). This incision can be extended superiorly under the SVC if the right PA is dissected off the roof of the LA, and inferiorly to the IVC. If exposure still poor the SVC can be transected so that the incision can extended more superiorly.

Dubost transeptal atriotomy: less commonly used. The bundle of His, and tricuspid annulus may be damaged by excessive retraction. It is helpful if previous AVR makes exposure via left atriotomy more difficult. A vertical incision is made in the lateral wall of the RA. The interatrial septum is opened at the fossa ovalis and extended to meet the right atriotomy. This is continued across the lateral aspect of the LA into the right SPV. Thin SPV may tear with retraction and are difficult to repair.

Guiraudon's (superior) vertical transeptal approach: RA opened longitudinally (appendage to IVC), fossa ovalis incised longitudinally, right atriotomy extended between AV groove and appendage to meet septal incision,

and dome of LA incised from the junction of these incision, away from the aortic root for 2–3cm. Gives good exposure of the anterior mitral annulus without excess retraction: tears in the LA dome may require patch closure and reinstutution of bypass to repair.

Closing the atriotomy

Use a continuous 3/0 Prolene in a single or double layer, starting at either end and tying in middle. *Pitfalls:* (1) closing off the RSPV orifice, (2) incomplete repair of the roof of the LA, (3) failure to evert the suture line predisposing to thrombus, (4) narrowing SVC, (5) sewing PA catheter into suture line. De-airing is described on 📖 p114.

Difficulty exposing the mitral valve

- Exposure is more difficult if there is a previous AVR, or LA is not dilated, e.g., acute MR from endocarditis: so think ahead.
- Exposure starts with set-up: hitch only right side of pericardium, incise caval pericardial reflections, in reop may need to free apex.
- Exposure helped by direct caval cannulation, lift up on caval snares.
- Place one retractor above the anterior annulus (not too deep or valve pushed posteriorly) and one at the posteromedial commissure, and tilt the table away from you slightly.
- Steadily increase retraction on each over several minutes.
- Make sure right heart and aortic root are fully decompressed.
- Assistant pressing down gently on aortic root with forceps handle can bring anterior annulus better into view.
- Extend incision superiorly onto dome LA ± inferiorly behind IVC ± towards fossa ovalis via intra-atrial septum (modified Dubost).
- Placing the commissural traction ± annular sutures where possible and pulling up usually brings all of the valve into view.

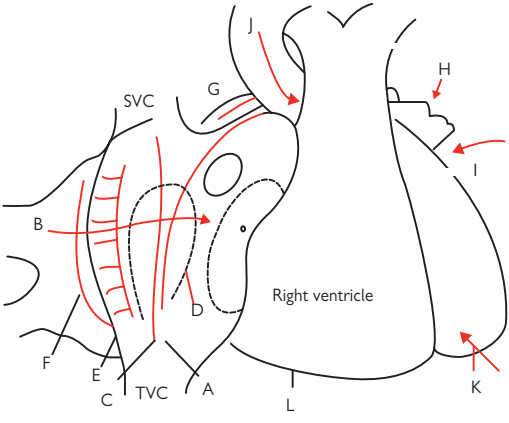


Fig. 8.12 Approaches to the mitral valve (for label key see text). Original drawing by Prasanna Simha MD. (See text for explanation of letters.)

Mitral valvotomy

Key facts

- Indicated for MS (usually rheumatic).
- Options are surgical or percutaneous (balloon mitral valvotomy BMV).
- Surgical approach either closed (without CPB) or open (requires CPB).
- Closed mitral valvotomy largely replaced by BMV in developed world.

Closed mitral valvotomy

First performed by Sir Henry Souttar in the UK in 1925. The basic principles of the operation remained unchanged for three decades until the advent of cardiopulmonary bypass enabled the surgeon to work on the mitral valve under direct vision, in a still, bloodless field.

Indications for closed mitral valvotomy

- Severe symptomatic MS.
- Patients should not have MR, and not have heavily calcified valves.

This technique has mostly been superseded by BMV for suitable patients in the West (and mitral replacement where BMV is contraindicated—see 📖 p407). Closed mitral valvotomy is only really still used in countries where rheumatic disease is prevalent and cost or lack of access to technology precludes BMV.

Technique of closed mitral valvotomy

Bypass is not required for closed valvotomy, which is performed via a left thoracotomy, with a finger inserted onto the valve via the LA appendage using a purse-string to prevent blood loss or air embolism, while a Tubbs dilator is passed into the LV via the apex and then opened within the orifice of the valve (Fig. 8.13a).

Results of closed mitral valvotomy

These are comparable to BMV with good improvement in valve area, low operative mortality, and freedom from stenosis for 5–10 years for most patients. 1/3 to 1/2 need repeat surgery within 15 years, 3/4 by 20 years.

Open mitral valvotomy

Indications

- Severe symptomatic MS—main lesion commissural fusion.

Technique of open mitral valvotomy (commissurotomy)

- Aorto-bicaval cannulation with antegrade cardioplegia via sternotomy or right mini-thoracotomy (📖 p110).
- Left atriotomy (📖 p404).
- Pull down anterior leaflet with forceps to delineate commissures.
- Incise the anterolateral commissure 3–4mm away from the annulus with a no. 11 blade, and extend it in the groove of the commissural fusion to the valve orifice (Fig. 8.13b). Staying 3–4mm away from the annulus prevents the commissure tearing and MR.
- Fused chordae present beneath the valve leaflets are separated with sharp dissection (chordal fenestration).
- The posteromedial commissure is treated in the same manner.

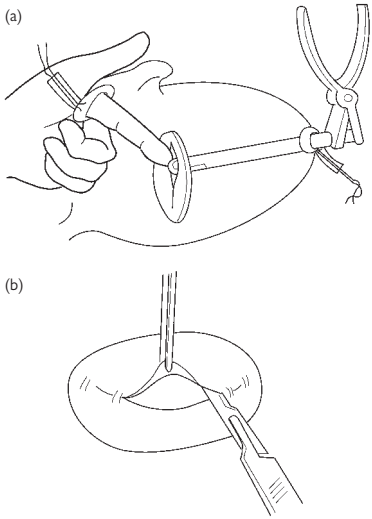


Fig. 8.13 (a) Closed mitral valvotomy. (b) Open mitral commissurotomy.

- Bulky calcium deposits may be removed from the leaflets and annulus with a bone nibbler, taking extreme care to avoid embolism.
- Satisfactory results can be obtained in the absence of calcification using either a finger or a Tubbs dilator to open up the valve.

Results of open mitral valvotomy

Comparable to BMV and open valvotomy in terms of valve area, operative mortality, and freedom from stenosis and reoperation in selected patients.

Balloon mitral valvotomy

Usually performed using transvenous approach with transseptal puncture. The initial results of percutaneous Inoue BMV versus surgical valvotomy are comparable, and both interventions have low rates of restenosis providing good functional outcomes at up to 3 years. At 3 years, however, valve areas in the BMV group are greater than in the surgery. In view of this, as well as the lower cost and morbidity of the percutaneous approach, percutaneous approach is recommended in patients without any contraindications,

Relative contraindications to BMV

- >2+ MR.
- LA thrombus.
- Wilkins score ≥ 8 (range 4–16), or Cormier group 3.
  - ↓leaflet mobility.
  - ↑leaflet/subvalvar thickening, calcification especially commissures.
- Severe concomitant CAD or AS requiring surgery.

Planning mitral valve repair

Key facts

- Match surgeon expertise to repair complexity to avoid replacements.
- Repair complexity depends on valve morphology: looking at the report is not enough—you *must* understand the images, before you start.
- History provides clues about etiology which helps predict morphology.
- MR is dynamic—3+MR on TTE in awake patient, may become trace MR on TEE in fasted patient under GA—always look at the pre-op films and do not rely on the pre-bypass TEE to plan repair.

Carpentier's classification

Carpentier classified MR as type I, normal leaflet motion; type II, excessive leaflet motion; type IIIa, restricted leaflet opening; type IIIb, restricted leaflet closure (see Table 8.3, p400). This is used with the anatomical division of each leaflet into three zones shown in Fig. 2.5. A reproducible system of repair is based on this nomenclature.

Planning the repair

- Review history for etiology—endocarditis, rheumatic fever, long patient long history murmur = Barlow's older patient, short history = FED.
- Review TTE and TEE in detail for leaflet dysfunction and lesions.
- Intraoperative inspection to confirm lesions.

Predicting repair complexity

- Valve analysis: clues from history, preoperative echo.

Lesions associated with more straightforward repair

- Type II *isolated* P2 from degenerative disease.
- Type IIIb from ischemic disease.

Lesions associated with more complex repair

- Multisegment or bileaflet prolapse, in Barlow's disease.
- Type IIIa lesion from rheumatic disease.
- Lesions related to infective endocarditis, e.g., leaflet perforation.
- Cleft mitral valve, primary SAM, previous failed mitral repair.

Set-up

Sternotomy or MICS (📖 p466) MICS less appropriate for endocarditis. Aorto-bicaval CPB, cardioplegia antegradely ± retrograde. The LV is vented directly through the mitral valve. TEE. Harvest pericardium if type IIIa rheumatic or endocarditis with perforation, fix in 0.08% glutaraldehyde for 20min, then rinse in 3 × 2min washes in saline.

Atriotomy

Usually paraseptal left atriotomy in Sondergaard's groove (📖 p404).

Assessment of the valve

This is the point at which a final decision can be made as to whether and what kind of repair is possible, or whether the valve must be replaced.

- Look for jet lesions on atrial wall to confirm direction regurgitant jet: anterior jet lesion = posterior leaflet prolapse or anterior restriction.
- Pressurize LV by injecting saline through the mitral valve using a bladder syringe with a flexible extension to see where the leaks are.
- Using two calibrated Carpentier hooks or nerve hooks of equal length examine each element of the valvular apparatus.
- First assess the annulus for dilatation (should be roughly same surface area as normal anterior leaflet).
- Next examine the chordae, distinguishing between normal primary chordae and abnormal stretched, ruptured, and fused chordae.
- Then, using the commissures as a fixed reference point, work systematically along each leaflet with the nerve hooks, assessing the location and degree of prolapse by gently lifting the edge of the leaflet with one hook and comparing the excursion with the fixed point of the nerve hook at the commissure.

Types of repair

These are outlined on 📖 p410. Common techniques illustrated in Fig. 8.14.

Testing the repair

This is done in several ways:

- Firstly by pressurizing LV with saline: there should be no residual leak.
- The line of leaflet coaptation should be symmetrical, parallel to and closest to the posterior annulus. It can be marked with ink, saline aspirated, and the area of coaptation beneath the ink checked—ideally >10mm for durability but <14mm to avoid SAM (📖 p190).
- Formal and definitive assessment can only be carried out by TEE once the heart is off bypass: up to 10% of patients will have more than mild MR and it is vital to go back on bypass to carry out further repair of valve replacement at this stage.

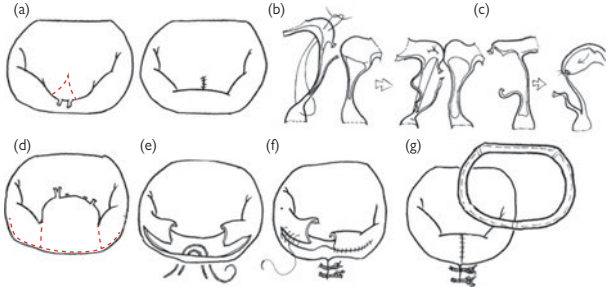


Fig. 8.14 Top row. Techniques for single-segment prolapse in small valve. (a) Triangular resection A2. (b) Gore-Tex loop. (c) Chordal transfer. Bottom row. Quadrangular resection of P2 prolapse and slidging plasty. (d) Resection lines. (e) Vertical compression sutures. (f) Sliding plasty of P1 and P3. (g) Final appearance of repair with remodelling annuloplasty ring superimposed, symmetrical closure line closer and parallel to posterior annulus.

Principles of mitral valve repair

Key facts

- Severe MR reduces normal life expectancy: repair is the only treatment that can restore this to normal: indications are listed on 📖 p402.
- Replacement has worse mortality and morbidity than repair in degenerative MR, but the data for ischemic MR is less clear.
- Mitral valve repair provides excellent hemodynamic and long-term results, without the need for anticoagulation.
- Carpentier's three principles of repair: (1) restore or preserve normal leaflet motion, (2) improve coaptation, (3) remodel the annulus.
- Techniques can be categorized by which part of the valvular apparatus they correct (Table 8.4).

Leaflets

- *Quadrangular resection ± sliding plasty*: this is used for posterior leaflet prolapse. A rectangular section of leaflet is excised from the area of maximal prolapse (Fig. 8.14). Several techniques allow the resected edges to be sutured together without tension: (1) sliding plasty, where the bases of the leaflet are divided from the annulus and reattached more centrally (Fig. 8.14g), (2) compression sutures (horizontal along most of the posterior annulus, and/or vertical just where the leaflet gap is (Fig. 8.14e).
- *Triangular resection:* this can be used for prolapse of the posterior or anterior leaflet (Fig. 8.14a). It is used where quadrangular resection would leave insufficient leaflet tissue for a competent repair. Historically anterior leaflet repairs have poorer durability than posterior.
- *Pericardial patch augmentation:* glutaraldehyde-fixed autologous pericardium (📖 p408) or bovine pericardium is used to extend posterior or anterior leaflet in rheumatic disease, and patch defects such as perforations in endocarditis or failed repairs.
- *Edge-to-edge:* Simple 4/0 proline suture used to approximate opposing leaflet edges, e.g., A2 to P2 (*Alfieri stitch*) in degenerative and ischemic MR. Poor durability without annuloplasty ring. Very useful at commissures to treat commissural prolapse (*Carpentier 'magic' suture*).
- *Commisurotomy*: (Fig. 8.13)

Chordae and papillary muscles

- *Loop technique:* Gore-Tex neochordae are fashioned between the papillary muscle and unsupported leaflet edge. Mohr's technique uses multiple pre-measured loops sewn to a pledget which is sewn to the base of a papillary muscle, and each loop is then sutured to the leaflet edge (Fig. 8.14b). Useful technique for any cause and location of leaflet prolapse.
- *Chordal cutting*: useful adjunct to ring annuloplasty for IIIb lesions improving leaflet mobility by cutting restrictive secondary chordae (Fig. 8.14c).
- *Chordal transfer and transposition:* the insertion of a normal secondary chord is excised, and re-attached to the flail segment leaflet (Fig. 8.14c). Transposition includes a segment of the leaflet.
- *Chordal shortening and reconstruction:* a variety of chordal shortening techniques have been described by Carpentier.
- *Papillary muscle reimplantation:* not ideal for ischemic papillary muscle rupture as the repair will not hold in necrotic muscle. The fibrous

tip of the ruptured head of papillary muscle is reattached to adjacent viable papillary muscle with mattress sutures.

Anulus

- *Reduction annuloplasty*: sutures or flexible band/ring used to reduce annular size to improve coaptation: thought to avoid SAM (📖 p191).
- *Remodeling annuloplasty*: usually complete, rigid or semirigid ring to reduce and remodel the annulus restoring systolic proportions (Fig. 8.15). Correct ring selection and sizing is critical to avoid MR and SAM, e.g.:
  - Type IIIb (ischemic)—use asymmetric down-sized rigid ischemic ring to address tethering predominantly in P2/P3 area, usually size 26–28.
  - Type II (degenerative)—use true-sized remodeling ring, pronounced saddle reduces risk of SAM post repair (📖 p191), usually size 34–40.

The ring is sized by extending the anterior leaflet with a valve hook, and comparing a valve sizer to the intercommissural distance, surface area, and height of the anterior leaflet. The ring is inserted with interrupted 2/0 or 3/0 non-absorbable braided sutures, using overlapping sutures on the posterior annulus if this is very dilated, and 1mm apart on the anterior annulus. The posterior annulus is reduced by making smaller bites in the valve-sewing ring than on the annulus, but the anterior annulus is not reduced to maintain the fixed intertrigonal distance.

Table 8.4 Common repair techniques by etiology and lesion

Etiology	Lesions	Repair techniques
Ischemic cardiomyopathy	Papillary muscle displacement, LV dilatation	Rigid down-sized ischemic ring (Fig. 8.15), ± chordal cutting
Degenerative disease	Annular dilatation	True-sized remodeling ring (Fig. 8.15)
(FED or Barlow's)	Chordal rupture or elongation	Chordal transfer, loops or Gore-Tex neo-chordae (Fig. 8.14b,c)
	Prolapse	Quadrangular resection ± sliding plasty (Fig. 8.14g) Triangular resection (Fig. 8.14a), Multiple loops, edge-to-edge
Rheumatic disease	Commissural fusion, leaflet calcification, chordal fusion	Pericardial patch augmentation, commisurotomy, chordal fenestration, ring



Fig. 8.15 Remodelling rings. (a) Carpentier–Edwards Physio. (b) IMR ETlogix Ring.

Mitral valve replacement

Indications for mitral valve replacement

- Repair is preferable to replacement in most cases of MR (📖 p402).
- Class 1: acute severe symptomatic MR, chronic severe MR *and* NYHA class II–IV *or* mild-moderate LV dysfunction. Also NYHA class III–IV moderate to severe MS if balloon valvotomy not possible.
- Class IIa: asymptomatic chronic severe MR with normal LV *if* new AF, *or* PA systolic >50mmHg at rest (60mmHg exercise). Also severe MS *if* PAP >60mmHg.
- Class IIb: chronic severe MR due to severe LV dysfunction despite optical medical therapy + biventricular pacing.

Set-up, bypass, and exposure are as described for MV repair (📖 p408).

Mitral valve replacement

The posterior leaflet and related subvalvular apparatus are preserved, where they will not interfere with prosthesis function. Chordal excision during MVR is associated with increases in postoperative LV end-systolic volume and wall stress, as well as a decrease in EF. Preserving the subvalvular apparatus optimizes ventricular function. Stenotic lesions are less amenable to this approach.

- To excise the valve begin with a 15 blade 2mm away from the annulus of the anterior leaflet at 12 o' clock where the leaflet tissue is usually most disease free (Fig. 8.16b).
- Continue the incision using either a McIndoe scissors or the 15 blade round to the commissural leaflet tissue at anterolateral and posteromedial commissures.
- Cut the very tips of the underlying papillary muscles. Cutting the chordae to the anterior leaflet risks scattering chordal fragments into the heart. The anterior leaflet can now be removed.
- In very elderly patients do not make overly aggressive efforts to remove subannular calcification: it may result in damage to the circumflex coronary artery, atrioventricular dehiscence or ventricular rupture and it is often possible to sew through the calcium and place a reasonable size prosthesis without undue difficulty.
- The valve is sized according to which sizer easily drops through the annulus into the ventricle taking care not to over-size or under-size the valve: a size 27 is almost always appropriate.
- The valve can be sewn using interrupted horizontal pledgetted everting or non-everting mattress sutures with a 2/0 braided non-absorbable suture, or using a semicontinuous technique with a 2/0 non-absorbable monofilament.
- Bites should incorporate the annulus, but deep bites may damage the circumflex artery (7–8 o'clock), coronary sinus (6–8 o'clock), NCC or LCC of aortic valve (11–1 o'clock), or AV node (2 o'clock) (Fig. 8.16a).
- Most surgeons orient bileaflet mechanical valves in the antianatomic position, and use everting mattress sutures to ensure that the valve leaflets move freely.

Chordal-sparing mitral valve replacement

To retain the anterior leaflet as well as the posterior leaflet, the anterior leaflet must be split centrally and folded laterally, or folded towards its base and incorporated in the suture line.

De-airing

This is described on 📖 p114. Frustrate the valve by inserting a 16 gauge Foley catheter. Do not lift the heart once off bypass: there is a risk of atrioventricular dehiscence (bright red blood pooling in pericardial well once bypass discontinued: requires resumption of bypass, removal of the prosthesis, and challenging patch repair of the rupture. Associated mortality is high.

Pitfalls in mitral valve insertion

- Over-sizing the valve leading to atrioventricular dehiscence, or subsequent difficulty inserting appropriate size AVR.
- Damaging the circumflex artery at 6–8 o'clock (Fig. 8.16a).
- Damaging the coronary sinus at 4–6 o'clock (Fig. 8.16a).
- Distorting the NCC or LCC sinus at 10–12 o'clock (Fig. 8.16a).
- Damaging the AV node at 1–2 o'clock (Fig. 8.16a).
- Failure to keep tension on the valve sutures while parachuting down leading to loops entrapping valve mechanism.
- AV dehiscence on lifting the heart to de-air or massage after MVR.
- Posts of bioprosthetic valve may cause LVOT obstruction, or rupture through LV wall (use low profile valve in small LV/elderly).

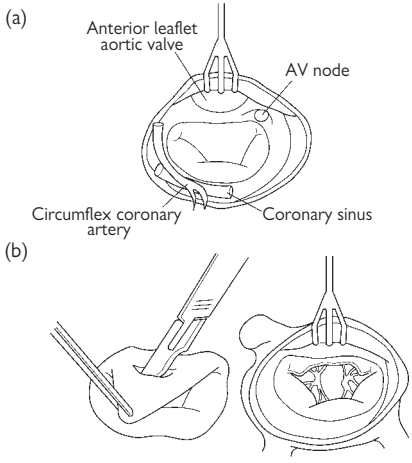


Fig. 8.16 (a) Structures at risk during MVR. (b) Excising the native valve.

Results of mitral surgery

Early mortality

- The mortality of isolated mitral ± tricuspid repair for degenerative disease in patients without pulmonary hypertension is ~1%.
- Higher operative mortality (2–5%) in patients with pulmonary hypertension, LV dysfunction, >80 years, ischemic MR, mitral replacement.

Late mortality

- Successful repair for degenerative MR restores life expectancy to that of an age-matched normal population.
- Ischemic MR is associated with much poorer survival: 1/3 of patients are dead within 5 years despite successful repair. LVEDD >65mm is associated with worse survival: <50% of patients alive at 5 years.
- Rheumatic patients are usually younger, and have good long-term survival with ~80–90% still alive 10 years after surgery.

Morbidity

These include the complications of cardiac surgery in general, i.e., re-sternotomy for bleeding/tamponade (1–2%), perioperative MI (1%), stroke (1%), deep sternal wound infection(<1%), prolonged ventilation (2%), and organ dysfunction (1%) are discussed in more detail on 📖 pp253–290. Complications particularly associated with mitral valve surgery include:

Failure of repair

Patient leaving the OR with >1+MR = *residual MR*. Patient left OR with no MR but develops it later = *recurrent MR*.

- Residual MR is due to a technical issue, e.g., incorrect choice of ring, failure to close cleft, inadequate treatment of prolapse.
- Recurrent MR may occur early (usually technical, e.g., dehiscence of leaflet closure, failure of chordal reconstruction but very occasionally endocarditis) or late (usually due to progression of underlying disease).
- In the midterm, failure of degenerative repair is most commonly technical, failure of rheumatic and ischemic repair tends to be due to disease progression.
- The late failure rate for degenerative disease is ~1–2% per year, and much lower than the failure rate for rheumatic and ischemic repair, reflecting the more progressive nature of the latter two etiologies.

Prosthetic endocarditis

Mechanical valves are at slightly higher risk of endocarditis in the first 3 months after surgery, the relative risks have evened out by 5 years. The cumulative risk of prosthetic valve endocarditis is ~1% at 1 year and 3% at 5 years post-MVR. Early endocarditis (within 3 months of surgery) is usually hospital-acquired: organisms are commonly staphylococci and Gram −ve cocci. Late prosthetic endocarditis usually results from a transient streptococcal bacteraemia.

Thromboembolism

The rate of valve thromboembolism is higher than in aortic prostheses because flow velocity across the valve is slower. It is roughly 2% per year, with adequate anticoagulation of mechanical valves, and 1% per year with bioprosthetic valves. May present as stroke or systemic embolization. The management is iv heparin. *Valve thrombosis* is rare. Presents with acute onset of extreme dyspnea and orthopnea, but may be insidious. Fluoroscopy or echo to visualize if one or both leaflets not moving. Acute thrombosis can be treated with thrombolytic, but more usually this is chronic process with acute exacerbation that requires replacement (normally with a bioprosthesis to avoid repeat episode). The thrombus is usually very organized and may be superimposed on pannus. Surgical thrombectomy tempting but ineffective as unable to clear pannus and thrombus easily from ventricular aspect of prosthesis.

Paraprosthetic leak

Paraprosthetic leak is uncommon: immediate or early may be due to needle holes (tiny, and disappear after protamine), inadequate decalcification of annulus, tear, or poor suture positioning. Late new paraprosthetic leak more likely due to endocarditis. Narrow, high-velocity jets are high risk for hemolytic anemia which may be disabling (requiring multiple transfusions). Broader jets may be hemodynamically significant especially if MR was not present preoperatively. The decision whether to treat depends on balancing disadvantages of fixing paravalvular leak (e.g., reclamping diseased aorta, additional ischemic time in sick heart, severe mitral annular calcification making any improvement unlikely, high-risk resternotomy) against the disadvantages of leaving it. Prosthetic valve endocarditis is a strong indication for surgery in this context.

Prosthesis failure and reoperation

In the mitral position the lifetime risk of reoperation for bioprosthetic failure is ~45% for a 50-year-old, and this falls by just over 10% for every additional 5 years of age at the time of implantation. At 20 years after surgery ~10–15% of patients with mechanical valves will have required reop valve replacement for infective endocarditis, pannus, or valve thrombosis.

Atrial fibrillation

Around 10–20% of patients undergoing mitral surgery have a history of pre-op AF and the incidence of post-op AF is ~40%. AF is associated with thromboembolic complications including stroke, heart failure, and decreased survival so aggressive treatment of patients with a history of pre-op AF with intra-op ablation (📖 p484) is recommended. These patients are routinely anticoagulated for 3 months after surgery, may be at increased risk of thrombus formation in the early post-op period, and are often pacemaker dependent for several days after surgery.

Tricuspid valve surgery

Key facts

- Functional TR secondary to left-sided heart disease, usually MR, is the commonest cause of tricuspid valve disease.
- Isolated tricuspid valve disease is uncommon: endocarditis from IV drug abuse is one of the main causes.
- Tricuspid disease associated with congenital defects (AV septal defect, pulmonary atresia, Ebstein anomaly) is discussed on 📖 pp636–637.

Pathophysiology and morphology

Functional tricuspid regurgitation: this commonly occurs secondary to left-sided valve lesions causing pulmonary hypertension and RV dilatation. When RV dilatation develops the tricuspid annulus dilates as it is not supported by a fibrous skeleton. This is most pronounced in the posterior and anterior part of the annulus, as the septal annulus is relatively fixed. Leaflet coaptation is reduced leading to TR. Observational data suggests that in patients with >2+TR, annular dilatation, or PAH, functional TR may not resolve with correction of the left-sided lesion, and should be repaired to avoid late severe TR, which develops in a minority of patients and is associated with high morbidity and operative mortality.

Tricuspid valve endocarditis: acute tricuspid valve endocarditis is increasing in incidence. It is usually associated with sepsis from long-term indwelling venous catheters or iv drug abuse. The causative organisms are *Pseudomonas aeruginosa* and *Staphylococcus aureus*.

Rheumatic tricuspid valve disease: most commonly this is TR predominantly due to annular dilatation secondary to severe MS. The tricuspid leaflets are often characterized by mild restriction and leaflet thickening.

Trauma: transvenous pacing wires can lead to leaflet perforation or more commonly scarring, causing TR. Traumatic TV rupture is a rare sequela of severe non-penetrating chest injury.

Carcinoid tricuspid valve disease: causes mixed tricuspid lesions (📖 p475).

Symptoms and signs

Tricuspid stenosis: the symptoms and natural history are determined by the dominant left-sided heart lesion. Tricuspid stenosis causes increased systemic venous pressure, resulting in hepatomegaly, ascites, and peripheral edema. A mid-diastolic murmur may be present.

Tricuspid regurgitation: the progression is slower than MR, and many patients are symptom-free for decades. The predominant clinical features are those of raised venous pressure. A pansystolic murmur may be present. Patients usually have moderate to severe RV failure by the time they present for surgery, as well as severe pulmonary hypertension, and moderate to severe liver dysfunction which is why isolated tricuspid valve surgery has such high mortality (~10–15%) with low cardiac output, respiratory failure, and sepsis the major problems postoperatively.

Indications for tricuspid valve surgery

- Class I: tricuspid repair for severe TR in patients having MV surgery.
- Class IIa: severe symptomatic primary TR.
- Class IIb: tricuspid repair for moderate TR ***or*** tricuspid annular dilatation in patients having mitral surgery.

Tricuspid valve surgery

Set-up and bypass: usually surgery is planned on the mitral valve at the same time: standard mitral set up is used. Venous drainage via bicaval cannulation is mandatory. Caval snares if no VAVD (📖 p103). If isolated TV then right thoracotomy, peripheral cannulation, beating heart, no clamp.

Right atriotomy: transverse right atriotomy, away from the AV groove, in the mid-atrium starting below the RA appendage, extending inferiorly (Fig. 8.17).

Tricuspid valve repair: the most common procedure is tricuspid annuloplasty, which may performed using a ring (rigid or flexible) or a running stitch around the annulus (DeVega) which is less durable. The tricuspid annulus tissue offers little support and sutures need to be placed and tied carefully to avoid tears, correcting TR by reducing the posterior and anterior annulus which are most prone to dilatation. The AV node is avoided by not placing sutures within the area of the triangle of Koch (Fig. 8.17). The repair can be tested by pressurizing the RV with saline instilled through the tricuspid valve while compressing the PA.

Tricuspid valve replacement: this is performed for failed repair, in endocarditis (where some surgeons advocate compete valvectomy) and carcinoid. A mechanical or bioprosthetic replacement can be used. The valve and subvalvular apparatus is excised, and the valve inserted using horizontal everting mattress sutures, with pledgets on the atrial side. It is important to avoid placing sutures at the apex of the triangle of Koch (Fig. 8.17) and sutures can be placed in the leaflet hinge instead in this area.

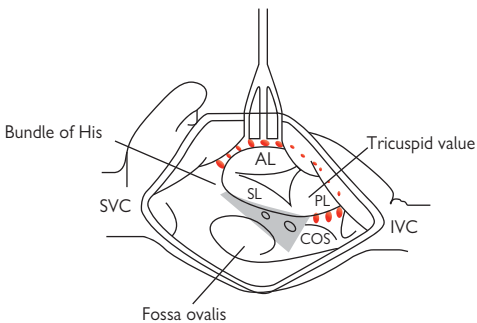


Fig. 8.17 Right atriotomy showing surgeon's view of tricuspid valve (AL, anterior; SL, septal; PL, posterior leaflet). Triangle of Koch bounded by coronary sinus (CoS), septal annulus, and tendon of Todaro inferiorly. Apex identifies conduction tissue. Dotted line represents suture placement for DeVega and partial ring annuloplasty.

Surgery for infective endocarditis

Key facts

- The pathophysiology of infective endocarditis is described in more detail on 📖 p80. This section deals with the surgical management.
- The Duke diagnostic criteria (Box 8.5) are 95% specific, with a 92% negative predictive value.
- Surgery indicated for structural valve lesions with evidence hemodynamic compromise or abscess formation.
- Optimal time to operate after embolic stroke is 7–14 days if possible.

Timing of surgery

Choosing the right time to operate can be difficult: the surgeon must define 'appropriate antibiotics', and balance need to optimize major comorbidity often including acute embolic stroke, with the fact that these patients are liable to deteriorate irreversibly. In most cases early surgical intervention is preferable to late: in case of preoperative embolic stroke delay 5–10 days if possible, if hemorrhagic stroke ideally longer.

Indications for surgery

- Class I: acute endocarditis + stenosis or regurgitation and CHF. Also acute endocarditis and AI or MR ***and*** PAH, ↑LAP, or ↑ LVEDP. Also fungal endocarditis. Also endocarditis complicated by heart block ***or*** annular ***or*** aortic abscess ***or*** fistulae between RA/RV/ LA and sinus of Valsalva ***or*** mitral leaflet perforation in aortic valve endocarditis. Also prosthetic valve endocarditis ***and*** dehiscence ***or*** worsening valve function ***or*** heart failure ***or*** abscess.
- Class IIa: native valve endocarditis ***and*** recurrent emboli and persistent vegetation ***despite*** appropriate antibiotics. Prosthetic valve endocarditis and bacteremia ***despite*** appropriate antibiotics.
- Class IIb: native valve endocarditis and mobile vegetation >10mm.

Box 8.5 Duke criteria for infective endocarditis

Definite: 2 major ***or*** 1 major + 3 minor ***or*** 5 minor.
Possible: 1 major + 1 minor ***or*** 3 minor.

Major criteria

- Blood culture (*Strep. viridans, Strep. bovis*, HACEK, *Staph. aureus,* enterococci with no primary focus from 2 separate cultures ***or*** persistent positive cultures >12h apart ***or*** all of 3 or majority of four separate cultures positive drawn over the space of >1h).
- Echo evidence: oscillating intracardiac mass in absence of alternative explanation ***or*** abscess ***or*** new regurgitant lesion ***or*** new partial dehiscence of prosthetic valve.

Minor criteria

- Predisposition/fever >38°C/emboli/blood culture or echo signs.

Technique of operation

The goals of surgery are:

- Remove infected tissue and drain abscesses.
- Reverse the hemodynamic abnormalities.
- Restore the cardiac and vascular architecture.

TEE is critical in planning surgery which may involve drainage of abscesses, debridement of necrotic tissue, closure of acquired defects such as VSD, annular abscess, fistulas, and aneurysms, in addition to valve repair and replacement.

Set-up

Median sternotomy (MICS rarely indicated because of possibility of more extensive involvement than identified pre-op), bicaval cannulation to permit access to the mitral and tricuspid valve if involved. Standard cardioplegia and venting techniques. If there are large mobile vegetations avoid handling the heart as much as possible, including placing retrograde, until the cross-clamp is on to reduce risk of emboli. Harvest pericardium for patch repair of mitral leaflet perforations (📖 p408).

Approach

Surgery for aortic endocarditis should involve examining the anterior leaflet of the mitral valve for lesions, and if necessary, a left atriotomy. Similarly surgery for mitral endocarditis should involve inspection of the aortic valve via an aortotomy if there is any suspicion of involvement.

- Send all excised tissue and debris to microbiology.
- There is no good evidence that homograft valve replacement reduces the risk of recurrent infective endocarditis.
- Aortic valve abscesses are commonest posterior to the membranous part of the interventricular septum, and under the left coronary ostia. Mitral valve abscesses are commonest in the P2 portion of the annulus.
- For small abscesses evacuation and debridement is relatively straightforward, but in extensive root abscesses involving the AV and ventriculo-arterial junctions debridement may cause dehiscence.
- Root replacement and patch repairs of AV discontinuity are indicated.
- Abscesses around the mitral valve should also be debrided with the same regard for AV discontinuity.
- In prosthetic valve endocarditis take extra care to debride all prosthetic material: pledgets may be difficult to see (particularly if on the ventricular side of the mitral annulus) but should be removed using rongeurs, along with any calcium not debrided first time round.
- Small vegetations can be debrided and perforations in the mitral valve may be repaired with autologous pericardium.
- Some surgeons swab the local area with solutions such as povidone-iodine, and change gloves and instruments once the valve is debrided. There is no evidence that this does any harm.
- Closure, de-airing, weaning from bypass, and assessing the valve function are carried out in the standard manner.

Combined valvular procedures

Key facts

- Multivalvar disease is usually rheumatic in origin. Infective endocarditis and connective tissue disease can also cause multivalvar disease.
- There are several possible sequences: should be tailored to individual patient, e.g., need to minimize ischemic time, complex mitral repair requiring saline testing versus replacement, small aortic root, CABG.

Pathophysiology

The pathophysiology is dictated by the dominant lesion, which varies:

- ***Aortic and mitral regurgitation:*** this imposes a large volume overload on the LV which undergoes marked hypertrophy and eventual dilatation.
- ***Aortic and mitral stenosis:*** this results in a small, hypertrophied, non-compliant LV, usually with well preserved LV EF.
- ***Mitral stenosis and aortic regurgitation:*** the MS is usually dominant. MS restricts LV filling, blunting the effect of AR on LV volume. Even in severe AR hyperdynamic circulation and dilated LV are absent.
- ***Aortic stenosis and mitral regurgitation:*** the AS worsens the degree of MR. The MR causes difficulty in assessing the severity of the AS because of reduced forward flow. Patients can decompensate rapidly.
- ***Mitral stenosis or regurgitation and tricuspid regurgitation:*** pulmonary hypertension is usually present (see 📖 p416 for more discussion).

Indications for surgery in moderate valve disease

Although the non-dominant lesion may be mild at the time of operation, these lesions are progressive. Rheumatic AS that was mild at the time of operation on concomitant rheumatic mitral lesion progresses to moderate to severe in 1/3 of patients within 5 years. The progression of AI is much slower and better tolerated, although distorting the aortic annulus by mitral annulolpasty ring or replacement may cause 2+AI to become severe. Regurgitant combined lesions tend to progress faster, and the prognosis in combined stenotic lesions is worse than in isolated stenotic lesions. The temptation to operate on multiple valves to avoid later surgery needs to be balanced against increased perioperative mortality and long-term morbidity of valve replacement. May need to inspect the valve in question at the time of surgery to make final decision.

Sequence of operation

Median sternotomy. If CABG is planned the LIMA and LSV are harvested before heparinization. Bicaval cannulation. The advantage of using retrograde cardioplegia is that it can be given without having to interrupt valve implantation. The patient is cooled to 32°C. There are several options for operative sequence after cross-clamping and arresting the heart:

- Sequence A: Allows saline testing of mitral repair, tricuspid and mitral exposure better before AVR in place:
  - Distal CABG anastomoses including LAD (bulldog on IMA).
  - TV repair.
  - Mitral +/– tricuspid repair then AVR.
  - Proximal anastomoses, de-air left heart, clamp off, close RA.

- Sequence B: Shortens clamp time, allows saline testing of mitral repair:
  - Distal CABG anastomoses including LAD (bulldog on IMA).
  - Mitral repair then AVR.
  - If single clamp technique perform proximals while giving hot-shot.
  - De-air left heart, clamp off.
  - Right atriotomy, TV repair beating heart, close right atriotomy.
  - Perform proximals with sidebiting clamp.
- Sequence C: No saline testing planned (mitral replacement not repair):
  - Distal CABG anastomoses, including LAD (bulldog on IMA).
  - Open aorta, excise and size aortic valve.
  - Mitral replacement.
  - Tricuspid repair.
  - Place aortic sutures and implant valve, close aortotomy, proximals.
  - Close LA, de-air left heart, clamp off, close RA.

Tips and pitfalls in combined valve surgery

- If AVR and CABG is planned, ensure aortic cannulation is as distal as possible so there is room for clamp, vent, proximals, and aortotomy: in a reop, axillary cannulation helps with this aspect (📖 p120).
- Exposure of the mitral and tricuspid is made more difficult by an aortic prosthesis, whereas a mitral prosthesis makes it more difficult to get a large prosthesis in the aortic position.
- Cannot saline test mitral from LA if aortotomy open.
- The tricuspid valve can be operated on with the clamp off and the heart beating, which reduces ischemic time (and allows sutures affecting the conduction tissue to be removed), but increases the risk of annular tears because of friable tissue and worse exposure.
- CABG distals should be performed first and checked for bleeding:
  - This reduces the need to pick the heart up once a mitral prosthesis is *in situ*, which can cause AV dehiscence.
  - Also allows cardioplegia to be given down the grafts.
- CABG proximals can be performed with the cross-clamp on (single-clamp technique) which minimizes stroke risk from multiple clamping, and avoids need to instrument the aortotomy closure, but adds to ischemic time.
- In multivalve reop surgery tissues are less compliant and it may be very difficult to get an adequate size aortic prosthesis, and occasionally mitral prosthesis in place:
  - Ensure both valve annulus completely debrided of calcium, pannus, pledgets and any remainder of the valve sewing ring.
  - Avoid placing the largest possible prosthesis in the mitral position as this will decrease the room available for an aortic prosthesis.
  - Consider the techniques for the small aortic root (📖 p392).
  - Occasionally aortic root enlargement with a patch extending down onto the dome of the LA roof and across the mitral annulus is required to allow implantation of appropriately-sized aortic and mitral prosthesis.

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